04 Complex Numbers

1. INTRODUCTION:

In the real number system, there is no solution to the equation $x^2 = -1$. In this lesson, we will study a new number system in which this equation does have a solution.

The backbone of this new number system is the number i, also known as the imaginary unit.

$\sqrt{-1} = i$

By taking multiples of this imaginary unit, we can create infinitely many more new numbers, like $3i$, $i\sqrt{3}$, and $-12i$. These are examples of **imaginary numbers.**

However, we can go even further than that and add real numbers and imaginary numbers, for example $2 + 7i$ and $3 - \sqrt{2}\,i$. These combinations are called **complex numbers.**

1.1 Integral Powers of $i = \sqrt{-1}$:

(i) $i^{4n} = 1, i^{4n+1} = i, i^{4n+2} = -1, i^{4n+3} = -i$ $(\forall\ n \in \mathbb{Z})$.

We know that $i^4 = 1 \Rightarrow \left(i^4\right)^n = 1\ \forall n \in \mathbb{Z}$

Let q be the quotient and r be the remainder, when 4 divides n i.e.

$n = 4q + r\ (0 \le r \le 3) \Rightarrow i^n = i^{4q+r} = i^{4q}.\, i^r = i^r$

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(ii) Sum of four consecutive integral powers of i is zero i.e.

$i^n + i^{n+1} + i^{n+2} + i^{n+3} = 0\ \ \forall n \in \mathbb{Z}$

1.2 Definition of Complex Numbers:

Complex numbers are defined as expressions of the form $x + iy$ where $x, y \in \mathbb{R}$ and $i = \sqrt{-1}$.

$z = x + iy$

Complex Number $(\mathbb{C})$

- $y = 0$ — Real Number $(\mathbb{R})$
- $y \neq 0$ — Non-Real Number $(\mathbb{C}\backslash\mathbb{R})$
  - $x = 0$ — Purely imaginary Number

Note :

(i) In $z = x + iy$, 'x' is called real part of z (denoted by Re(z)) and 'y' is called imaginary part of z (denoted by Im(z))

(ii) The set $\mathbb{R}$ of real numbers is a proper subset of the set of Complex Numbers $(\mathbb{C})$. Observe:

$\mathbb{N} \subset \mathbb{N}_0 \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$.

(iii) The number (z) is called purely imaginary if it is equal to iy with $y \neq 0$.

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Illustration 1

Find the value of $i^{57} + \frac{1}{i^{125}}$

Solution:

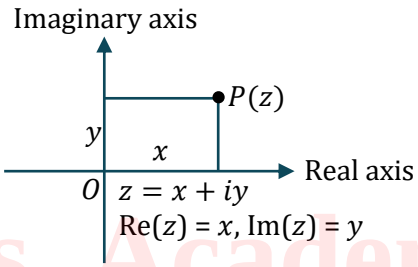
$$i^{57} + \frac{1}{i^{125}} = i^{56}.\, i + \frac{1}{i^{124}.i}$$

$$= \left(i^4\right)^{14} i + \frac{1}{\left(i^4\right)^{31} i}$$

$$= i + \frac{1}{i} = i + \frac{i}{i^2} = i - i = 0$$

1.3 Argand Diagram:

Master Argand had done a systematic study on complex numbers and represented every complex number $z = x + iy$ as a set of ordered pair (x, y) on a plane called complex plane (Argand Diagram) containing two perpendicular axes. Horizontal axis is known as Real axis and vertical axis is known as Imaginary axis.



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All complex numbers lying on the real axis are called real numbers and all complex numbers lying on imaginary axis other than origin are called purely imaginary numbers.

2. ALGEBRAIC OPERATIONS:

Fundamental operations with complex numbers are defined as follows : $(a,b,c,d \in \mathbb{R})$

(a) Addition : $(a + bi) + (c + di) = (a + c) + (b + d)i$

(b) Subtraction : $(a + bi) - (c + di) = (a - c) + (b - d)i$

(c) Multiplication : $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$

(d) Division : $\frac{a+bi}{c+di} = \frac{a+bi}{c+di}.\frac{c-di}{c-di} = \frac{ac+bd}{c^2+d^2} + \frac{bc-ad}{c^2+d^2}i$

Note : The algebraic operations on complex numbers are similar to those on real numbers treating i as a monomial.

Illustration 2

Find the value of $\left(\frac{1+i}{1-i}\right)^2 + \left(\frac{1-i}{1+i}\right)^2$.

Solution:

$$\left(\frac{1+i}{1-i}\right)^2 + \left(\frac{1-i}{1+i}\right)^2 = \left[\frac{(1+i)(1+i)}{(1+i)(1-i)}\right]^2 + \left[\frac{(1-i)(1-i)}{(1+i)(1-i)}\right]^2 = \left[\frac{1+i^2+2i}{1-i^2}\right]^2 + \left[\frac{1+i^2-2i}{1-i^2}\right]^2$$

$$= \left[\frac{1-1+2i}{1+1}\right]^2 + \left[\frac{1-1-2i}{1+1}\right]^2 = i^2 + (-i)^2 = -1 - 1 = -2$$

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Illustration 3

Find general value(s) of θ, if $z = \frac{3+2i\sin\theta}{1-2i\sin\theta}$ is purely imaginary.

Solution:

$$\frac{3+2i\sin\theta}{1-2i\sin\theta} = \frac{(3+2i\sin\theta)}{(1-2i\sin\theta)} \times \frac{(1+2i\sin\theta)}{(1+2i\sin\theta)} = \frac{\left(3-4\sin^2\theta\right)+i\left(8\sin\theta\right)}{\left(1+4\sin^2\theta\right)}$$

$$\text{Re}\left(\frac{3+2i\sin\theta}{1-2i\sin\theta}\right) = 0 \Rightarrow \frac{3-4\sin^2\theta}{1+4\sin^2\theta} = 0 \Rightarrow 3 - 4\sin^2\theta = 0$$

$$\Rightarrow \sin^2\theta = \left(\frac{\sqrt{3}}{2}\right)^2 = \left(\sin\frac{\pi}{3}\right)^2$$

$$\Rightarrow \theta = n\pi \pm \frac{\pi}{3};\ n \in I$$

2.1 Equality In Complex Numbers:

Two complex numbers $z_1 = a_1 + ib_1$ and $z_2 = a_2 + ib_2$; $a_i, b_i \in \mathbb{R}$ for $i = 1,2$ are equal if and only if their real and imaginary parts are respectively equal. i.e.

$a_1 + ib_1 = a_2 + ib_2 \Leftrightarrow a_1 = a_2$ and $b_1 = b_2$.

Note :

(i) Inequality in non-real complex numbers are never talked. For $a, b, c, d \in \mathbb{R}$, $a + ib > c + id$ will be meaningful only if $b = d = 0$ and statements like $1 + i > 0$, $4 + 2i < 2 + 4i$ are meaningless.

(ii) In real numbers, $a^2 + b^2 = 0 \Rightarrow a = 0 = b$ but in complex numbers, $z_1^2 + z_2^2 = 0$ does not imply $z_1 = z_2 = 0$. For e.g. $5^2 + (5i)^2 = 0$

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Illustration 4

Find the values of x and y satisfying the equation $\frac{(1+i)x-2i}{3+i} + \frac{(2-3i)y+i}{3-i} = i$.

Solution:

$$\frac{(1+i)x-2i}{3+i} + \frac{(2-3i)y+i}{3-i} = i \Rightarrow (4 + 2i)\,x + (9 - 7i)\,y - 3i - 3 = 10i$$

Equating real and imaginary parts, we get $2x - 7y = 13$ and $4x + 9y = 3$.

Solving, we get $x = 3$ and $y = -1$.

Illustration 5

If $a^2 + b^2 = 1$, then prove that $\frac{1+b+ia}{1+b-ia} = b + ia$; $a, b \in \mathbb{R}$.

Solution:

Given that $a^2 + b^2 = 1$

$$\therefore \frac{1+b+ia}{1+b-ia} = \frac{(1+b+ia)(1+b+ia)}{(1+b-ia)(1+b+ia)}$$

$$= \frac{(1+b)^2 - a^2 + 2ai(1+b)}{1+b^2+2b+a^2} = \frac{\left(1-a^2\right)+2b+b^2+2ia(1+b)}{2(1+b)} = \frac{2b^2+2b+2ia(1+b)}{2(1+b)} = b + ia$$

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2.2 Square Root of A Complex Number:

Let $z = x + iy; x, y \in \mathbb{R}$ be square root of complex number $\omega = a + ib; a, b \in \mathbb{R}$ i.e.

$(x+iy)^2 = a + ib \Rightarrow x^2 - y^2 + i2xy = a + ib$

Comparing real and imaginary parts. we get

$x^2 - y^2 = a$ and $2xy = b \Rightarrow a^2 + b^2 = (x^2 + y^2)^2$

Now, solve $\begin{Bmatrix} x^2 + y^2 = \sqrt{a^2 + b^2} \\ x^2 - y^2 = a \\ 2xy = b \end{Bmatrix}$ and find x, y.

Obverse the following illustration

Illustration 6

Find the square root of $7 + 24i$.

Solution:

Let $7 + 24i = (a + ib)^2 \Rightarrow a^2 - b^2 + 2iab = 7 + 24i$

Comparing real and imaginary parts, we get $a^2 - b^2 = 7$ and $2ab = 24$

By solving these two equations, we get $a = \pm 4, b = \pm 3$

Also $ab = 12 \Rightarrow a, b$ are of same sign

$\Rightarrow a + ib = \pm(4 + 3i)$

Illustration 7

If $x = -5 + 2\sqrt{-4}$, find the value of $x^4 + 9x^3 + 35x^2 - x + 4$.

Solution:

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We have, $x = -5 + 2\sqrt{-4}$

$\Rightarrow x + 5 = 4i \Rightarrow (x + 5)^2 = 16i^2$

$\Rightarrow x^2 + 10x + 25 = -16 \Rightarrow x^2 + 10x + 41 = 0$

Dividing the required value by $x^2 + 10x + 41$, we get

$x^4 + 9x^3 + 35x^2 - x + 4 = x^2(x^2 + 10x + 41) - x(x^2 + 10x + 41) + 4(x^2 + 10x + 41) - 160$

$= x^2(0) - x(0) + 4(0) - 160 = -160$

3. CONJUGATE OF A COMPLEX NUMBER:

Conjugate of a complex number, z is obtained by changing the sign of its imaginary part and is denoted by $\bar{z}$ i.e.

If $z = a + ib; a, b \in \mathbb{R}$ then $\bar{z} = a - ib$.

3.1 Properties of Conjugate:

(a) $z + \bar{z} = 2 \text{ Re } (z)$

(b) $z - \bar{z} = 2i \text{ Im } (z)$

(c) $\overline{(\bar{z})} = z$

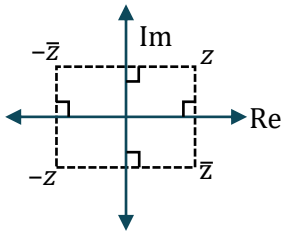
(d) $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$

(e) $\overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2$

(f) $\overline{z_1 z_2} = \bar{z}_1 . \bar{z}_2$. In general $\overline{z_1 z_2 \ldots z_n} = \bar{z}_1 . \bar{z}_2 \ldots \bar{z}_n$

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(g) $\overline{\left(\frac{z_1}{z_2}\right)}=\frac{\bar{z}_1}{\bar{z}_2}$; $z_2\neq 0$



(h) If $z=a+bi;a,b\in\mathbb{R}$, then $z\cdot\bar{z}=a^2+b^2$

$z\cdot\bar{z}=(a+bi)(a-bi)=a^2-i^2b^2=a^2+b^2$

(i) $z-\bar{z}=0\Leftrightarrow$ z is real

(j) $z+\bar{z}=0\Leftrightarrow z$ is purely imaginary

(k) If z lies in the 1st quadrant, then $\bar{z}$ lies in the 4th quadrant and $-\bar{z}$ lies in the 2nd quadrant.

Illustration 8

For any three complex numbers z_1, z_2 and z_3 prove that :

$z_1\,\mathrm{Im}(\bar{z}_2z_3)+z_2\,\mathrm{Im}(\bar{z}_3z_1)+z_3\,\mathrm{Im}(\bar{z}_1z_2)=0$

Solution:

Using $\mathrm{Im}(z)=\frac{1}{2i}(z-\bar{z})$, we get

L.H.S. = $z_1\,\mathrm{Im}(\bar{z}_2z_3)+z_2\,\mathrm{Im}(\bar{z}_3z_1)+z_3\,\mathrm{Im}(\bar{z}_1z_2)$

$=\frac{1}{2i}\left(z_1(\bar{z}_2z_3-z_2\bar{z}_3)\right)+z_2(\bar{z}_3z_1-z_3\bar{z}_1)+z_3(\bar{z}_1z_2-z_1\bar{z}_2)$

$=\frac{1}{2i}(z_1\bar{z}_2z_3-z_1z_2\bar{z}_3+z_2\bar{z}_3z_1-z_2z_3\bar{z}_1+z_3\bar{z}_1z_2-z_3z_1\bar{z}_2)=0$ = R.H.S.

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Illustration 9

Consider a quadratic equation $az^2 + bz + c = 0$ where a, b, c are complex numbers. Find the condition that the equation has one purely real root.

Solution:

Let z_1 be the real root of the given equation $\Rightarrow z_1=\bar{z}_1$ and $az_1^2+bz_1+c=0$ ...(1)

$\Rightarrow\overline{az_1^2+bz_1+c}=\bar{0}\Rightarrow\overline{az_1^2}+\overline{bz_1}+\bar{c}=0\Rightarrow\bar{a}z_1^2+\bar{b}z_1+\bar{c}=0$ ...(2)

Now (1) and (2) must have one common root

Required condition is $(b\bar{c}-c\bar{b})(a\bar{b}-\bar{a}b)=(c\bar{a}-a\bar{c})^2$

Illustration 10

Consider a quadratic equation $az^2 + bz + c = 0$ where a, b, c are complex numbers. Find the condition such that both the roots of given quadratic are purely real.

Solution:

Let z_1 and z_2 be two purely real roots then $\bar{z}_1=z_1,\bar{z}_2=z_2$

Hence, in this case $az^2 + bz + c = 0$ ...(1)

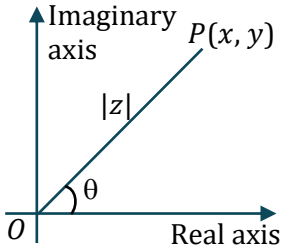
$\Rightarrow\bar{a}\,\bar{z}^2+\bar{b}\bar{z}+\bar{c}=0\Rightarrow\bar{a}z^2+\bar{b}z+\bar{c}=0$ ...(2)

Equation (1) and (2) must have two common roots

$\Rightarrow\frac{a}{\bar{a}}=\frac{b}{\bar{b}}=\frac{c}{\bar{c}}$ and $\left(\frac{b}{a}\right)^2-4\left(\frac{c}{a}\right)\geq 0$

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4. MODULUS OF A COMPLEX NUMBER:



If P denotes complex number $z = x + iy$, then the length OP is called modulus of complex number z. It is denoted by $|z|$.

$OP = |z| = \sqrt{x^2 + y^2}$

4.1 Properties of Modulus:

(a) $|z| \geq 0$

(b) $|z| \geq \text{Re}(z)$

(c) $|z| \geq \text{Im}(z)$

(d) $|z| = |\bar{z}| = |-z| = |-\bar{z}|$

(e) $z \cdot \bar{z} = |z|^2$

(f) $|z_1 z_2| = |z_1|.|z_2|$. In general $|z_1 z_2 \ldots z_n| = |z_1|.|z_2|\ldots|z_n|$

(g) $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}, z_2 \neq 0$

(h) $|z^n| = |z|^n, n \in \mathbb{Z}$

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$|z_1 + z_2|^2 = (z_1 + z_2)\overline{(z_1 + z_2)} = (z_1 + z_2)(\bar{z}_1 + \bar{z}_2)$

(i)
$$= |z_1|^2 + |z_2|^2 + z_1\bar{z}_2 + \bar{z}_1 z_2$$
$$= |z_1|^2 + |z_2|^2 + 2\,\text{Re}(z_1\bar{z}_2)$$

(j) $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2\left(|z_1|^2 + |z_2|^2\right)$

(k) Unlike real numbers, $|z| = \begin{cases} z, & z \geq 0 \\ -z, & z < 0 \end{cases}$ is not correct (generally).

Illustration 11

If $|z_1| = |z_2| = |z_3| = 1$, prove that $|z_1 + z_2 + z_3| = \left|\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right|$.

Solution:

We know that $|z| = |\bar{z}|$

$\Rightarrow |z_1 + z_2 + z_3| = |\bar{z}_1 + \bar{z}_2 + \bar{z}_3|$

$$= \left|\frac{\bar{z}_1.z_1}{z_1} + \frac{\bar{z}_2.z_2}{z_2} + \frac{\bar{z}_3.z_3}{z_3}\right| = \left|\frac{|z_1|^2}{z_1} + \frac{|z_2|^2}{z_2} + \frac{|z_3|^2}{z_3}\right|$$

$$= \left|\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right| \qquad (\because |z_1|^2 = |z_2|^2 = |z_3|^2 = 1)$$

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Illustration 12

Prove that $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2)$ is true for all $z_1, z_2 \in \mathbb{C}$.

Solution:

$|z_1 + z_2|^2 = (z_1 + z_2)(\bar{z}_1 + \bar{z}_2) = z_1\bar{z}_1 + z_2\bar{z}_2 + z_1\bar{z}_2 + z_2\bar{z}_1$

$|z_1 - z_2|^2 = (z_1 - z_2)(\bar{z}_1 - \bar{z}_2) = z_1\bar{z}_1 + z_2\bar{z}_2 - z_1\bar{z}_2 - z_2\bar{z}_1$

Adding, $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(z_1\bar{z}_1 + z_2\bar{z}_2) = 2\left(|z_1|^2 + |z_2|^2\right)$

Illustration 13

For complex numbers z and w, prove that $|z|^2 w - |w|^2 z = z - w \Leftrightarrow z = w$ or $z\bar{w} = 1$.

Solution:

$\frac{z}{w} = \frac{|z|^2 + 1}{|w|^2 + 1} \Rightarrow \frac{z}{w}$ is purely real $\Rightarrow \frac{z}{w} = \overline{\left(\frac{z}{w}\right)} \Rightarrow z\bar{w} = \bar{z}w$ ...(i)

$\therefore\ |z|^2 w - |w|^2 z = z - w \Rightarrow z\bar{z}w - w\bar{w}z - z - w$

$\Rightarrow z\bar{w}(z - w) = z - w$ [using $z\bar{w} = \bar{z}w$]

$\Rightarrow (z\bar{w} - 1)(z - w) = 0$ Hence proved

Similarly, its converse can also be proved.

Illustration 14

Prove that $\left|\alpha - \sqrt{\alpha^2 - \beta^2}\right| + \left|\alpha + \sqrt{\alpha^2 + \beta^2}\right| = |\alpha + \beta| + |\alpha - \beta|$ is true for all $\alpha, \beta \in \mathbb{C}$.

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Solution:

$\left|\alpha - \sqrt{\alpha^2 - \beta^2}\right| + \left|\alpha + \sqrt{\alpha^2 - \beta^2}\right| = \frac{1}{2}\left\{\left|2\alpha - 2\sqrt{\alpha^2 - \beta^2}\right| + \left|2\alpha + 2\sqrt{\alpha^2 - \beta^2}\right|\right\}$

$= \frac{1}{2}\left\{\left|\alpha + \beta + \alpha - \beta - 2\sqrt{\alpha^2 - \beta^2}\right| + \left|\alpha + \beta + \alpha - \beta + 2\sqrt{\alpha^2 - \beta^2}\right|\right\}$

$= \frac{1}{2}\left\{\left|\sqrt{\alpha + \beta} - \sqrt{\alpha - \beta}\right|^2 + \left|\sqrt{\alpha + \beta} + \sqrt{\alpha - \beta}\right|^2\right\}$

$= \left|\sqrt{\alpha + \beta}\right|^2 + \left|\sqrt{\alpha - \beta}\right|^2 = |\alpha + \beta| + |\alpha - \beta|$

Illustration 15

Let z_1 and z_2 are two complex numbers such that $\frac{z_1 - 2z_2}{2 - z_1\bar{z}_2}$ is unimodular (whose modulus is one), while z_2 is not unimodular. Find $|z_1|$.

Solution:

Here $\left|\frac{z_1 - 2z_2}{2 - z_1\bar{z}_2}\right| = 1 \Rightarrow |z_1 - 2z_2| = |2 - z_1\overline{z_2}|$

$\Rightarrow |z_1 - 2z_2|^2 = |2 - z_1\overline{z_2}|^2$

$\Rightarrow (z_1 - 2z_2)(\overline{z_1} - 2\overline{z_2}) = (2 - z_1\overline{z_2})(2 - \overline{z_1}z_2)$

$\Rightarrow |z_1|^2 - 2z_1\bar{z}_2 - 2z_2\bar{z}_1 + 4|z_2|^2 = 4 - 2\bar{z}_1z_2 - 2z_1\bar{z}_2 + |z_1z_2|^2$

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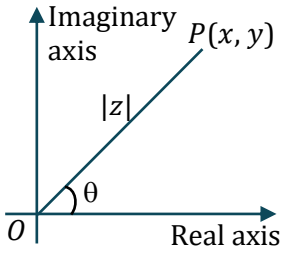
$\Rightarrow |z_1|^2 + 4|z_2|^2 = 4 + |z_1|^2 |z_2|^2$

$\Rightarrow \left(|z_1|^2 - 4\right)\left(1 - |z_2|^2\right) = 0$

But $|z_2| \neq 1$ (given) $\Rightarrow |z_1|^2 = 4$

Hence, $|z_1| = 2$.

5. ARGUMENT/AMPLITUDE OF A COMPLEX NUMBER:



If P denotes complex number $z = x + iy$ and if OP makes an angle θ with positive real axis, then θ is called one of the arguments of z.

Note :

(i) Argument of a complex number is a many valued function. If θ is the argument of a complex number, then $2n\pi + \theta$; $n \in Z$ will also be the argument of that complex number. Any two arguments of a complex number differ by $2n\pi$; $n \in Z$.

(ii) The unique value of θ such that $0 < \theta \leq 2\pi$ is called **least positive argument.**

(iii) The unique value of θ such that $0 \leq \theta < 2\pi$ is called **least non-negative argument.**

(iv) The unique value of θ such that $-\pi < \theta \leq \pi$ is called **Principal value of the argument.**

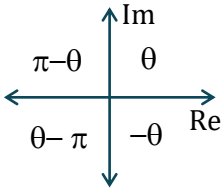
(v) By specifying the modulus and argument, a complex number is defined completely. Argument impart direction and modulus impart distance from origin.

(vi) For the complex number $0 + i0$, the argument is not defined and this is the only complex number which is given by its modulus only.

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5.1 Finding Principle Value of Argument:

Principal argument of a complex number $z = x + iy; x, y \in R - \{0\}$ can be found out using method given below (i.e. when z lies in any one of the quadrants):



Step -1 : Find $\theta = \tan^{-1}\left|\frac{y}{x}\right|$ and identify the quadrant of z.

Step -2 : Use given figure to find out the principal argument according as the point lies in respective quadrant.

Following points must be noted for z lying on axes :

(Here amp(z) denotes the principle value of argument.)

(a) If z lies on negative side of real axis, amp(z) = π.

(b) If z lies on positive side of real axis, amp(z) = 0.

(c) If z lies on the positive side of imaginary axis, amp(z) = $\frac{\pi}{2}$

(d) If z lies on the negative side of imaginary axis, amp(z) = $-\frac{\pi}{2}$

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5.2 Properties of Argument/Amplitude:

Let amp(z) denote the principle value of argument. Then amp(z) satisfies the following properties:

(a) amp $(z_1.z_2)$ = amp z_1 + amp z_2 + $2k\pi$; $k \in Z$

(b) $amp\left(\frac{z_1}{z_2}\right)$ = amp z_1 + amp z_2 + $2k\pi$; $k \in Z$

(c) amp(z^n) = n amp(z) + $2k\pi$; $n,k \in Z$

(d) amp($\bar{z}$) = – amp(z)

Note : Proper value of $k(\in Z)$ must be chosen so that RHS in properties (a), (b) and (c) lies in the interval $(-\pi,\pi]$

Illustration 16

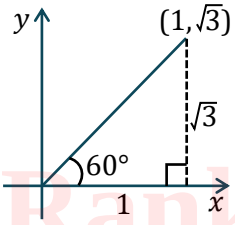
Find the modulus, argument, principal value of argument, least positive argument of following complex numbers

(A) $1 + i\sqrt{3}$ (B) $-1 + i\sqrt{3}$ (C) $1 - i\sqrt{3}$ (D) $-1 - i\sqrt{3}$

Solution:

Let amp(z) denote the principle value of argument.

(a) For $z = 1 + i\sqrt{3}$



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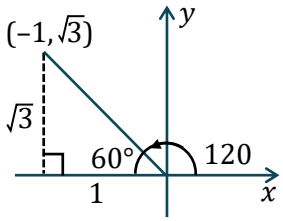
$|z| = \sqrt{1^2 + (\sqrt{3})^2} = 2$

arg (z) = $2n\pi + \frac{\pi}{3}$, $n \in I$

Least positive argument is $\frac{\pi}{3}$

If the point is lying in first or second quadrant then amp(z) is taken in anticlockwise direction.

In this case amp(z) = $\frac{\pi}{3}$



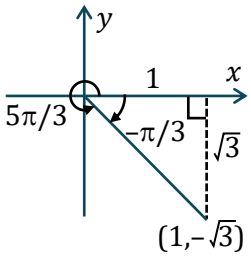
(b) For $z = -1 + i\sqrt{3}$

$|z| = 2$

arg (z) = $2n\pi + \frac{2\pi}{3}$, $n \in I$

Least positive argument = $\frac{2\pi}{3}$

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amp$(z) = \frac{2\pi}{3}$

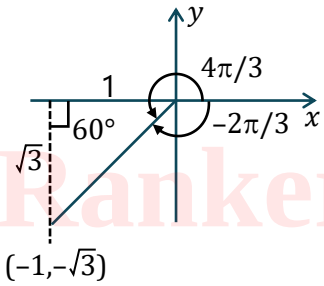
(c) For $z = 1 - i\sqrt{3}$

$|z| = 2$

arg $(z) = 2n\pi - \frac{\pi}{3}, n \in I$

Least positive argument = $\frac{5\pi}{3}$

If the point lies in third or fourth quadrant then consider amp(z) in clockwise direction.



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In this case amp$(z) = -\frac{\pi}{3}$

(d) For z = $-1 - i\sqrt{3}$

$|z| = 2$

arg $(z) = 2n\pi - \frac{2\pi}{3}, n \in I$

Least positive argument = $\frac{4\pi}{3}$

amp$(z) = -\frac{2\pi}{3}$

Illustration 17

Find amp z and $|z|$ if $z = \left[\frac{(3+4i)(1+i)(1+\sqrt{3}i)}{(1-i)(4-3i)(2i)}\right]^2$.

Solution:

Let amp(z) denote the principle value of argument.

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Method-I :

$\text{amp } z = 2\left[\text{amp}(3+4i)+\text{amp}(1+i)+\text{amp}(1+\sqrt{3}i)-\text{amp}(1-i)-\text{amp}(4-3i)-\text{amp}(2i)\right]+2k\pi$

where $k \in I$ and k chosen so that amp z lies in $(-\pi,\pi]$.

$\Rightarrow \text{amp}\, z = 2\left[\tan^{-1}\frac{4}{3}+\frac{\pi}{4}+\frac{\pi}{3}-\left(-\frac{\pi}{4}\right)-\tan^{-1}\left(-\frac{3}{4}\right)-\frac{\pi}{2}\right]+2k\pi$

$\Rightarrow \text{amp } z = 2\left[\tan^{-1}\frac{4}{3}+\cot^{-1}\frac{4}{3}+\frac{\pi}{3}\right]+2k\pi \Rightarrow \text{amp}\, z = 2\left[\frac{\pi}{2}+\frac{\pi}{3}\right]+2k\pi$

$\Rightarrow \text{amp } z = -\frac{\pi}{3}$ [at $k = -1$]

Also, $|z| = \left|\frac{(3+4i)(1+i)\left(1+\sqrt{3}i\right)}{(1-i)(4-3i)(2i)}\right|^2 \Rightarrow |z| = \left(\frac{|3+4i||1+i|\left|1+\sqrt{3}i\right|}{|1-i||4-3i||2i|}\right)^2$

$\Rightarrow |z| = \left(\frac{5\times\sqrt{2}\times 2}{\sqrt{2}\times 5\times 2}\right)^2 = 1$

Method-II :

$z = \left[\frac{(3+4i)(1+i)\left(1+\sqrt{3}i\right)}{(1-i)(4-3i)(2i)}\right]^2 \Rightarrow z = \left[-\frac{\sqrt{3}+i}{2}\right]^2 \Rightarrow z = \frac{2-2\sqrt{3}i}{4} = \frac{1}{2}-\frac{\sqrt{3}i}{2}$

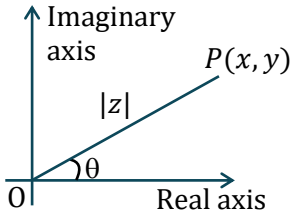
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Hence $|z| = 1$, amp$(z) = -\frac{\pi}{3}$

6. REPRESENTATION OF A COMPLEX IN DIFFERENT FORMS:

6.1 Cartesian Form (Geometrical Representation):

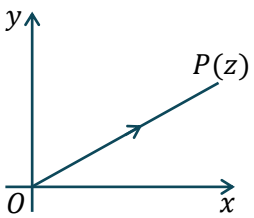
Every complex number $z = x + i\,y$ can be represented by a point on the cartesian plane known as complex plane by the ordered pair (x, y). There exists a one-one correspondence between the points of the plane and the members of the set of complex numbers.



For $z = x + iy$; $|z| = \sqrt{x^2+y^2}$; $\bar{z} = x - iy$

6.2 Vectorial Representation:

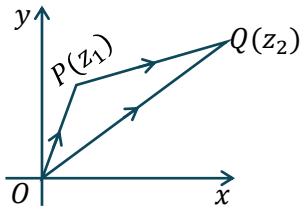
In complex number every point can be represented in terms of position vector. If the point P represents the complex number z then, $\overrightarrow{OP} = z$ & $|\overrightarrow{OP}| = |z|$.



Note: If $P(z_1)$ & $Q(z_2)$ be two complex numbers on argand plane then $\overrightarrow{PQ}$ represents complex number $z_2 - z_1$.

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6.3 Trigonometric/Polar Representation:



$z = r(\cos\theta + i\sin\theta)$ where $|z| = r$; $\arg z = \theta$; $\bar{z} = r(\cos\theta - i\sin\theta)$

Note : $\cos\theta + i\sin\theta$ is also written as cis (θ).

6.4 Exponential Representation:

Let z be a complex number such that $|z| = r$ and $\arg z = \theta$, then $z = r.e^{i\theta}$; $e^{i\theta} = \cos\theta + i\sin\theta$.

Note :

The formula $\boldsymbol{e^{i\theta} = \cos\theta + i\sin\theta}$ is called Euler's formula. It was introduced by Euler in 1748.

7. DE-MOIVRE'S THEOREM:

For any real θ and positive integer n, we have

$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$.

Illustration 18

Express the following complex numbers in polar and exponential form:

(i) $\dfrac{1+3i}{1-2i}$ (ii) $\dfrac{i-1}{\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}}$

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Solution:

(i) Let $z = \dfrac{1+3i}{1-2i} = \dfrac{1+3i}{1-2i}\times\dfrac{1+2i}{1+2i} = -1+i$

$\Rightarrow |z| = \sqrt{(-1)^2+1^2} = \sqrt{2}$ and $\theta = \text{amp}(z) = \pi - \alpha = \pi - \dfrac{\pi}{4} = \dfrac{3\pi}{4}$

Hence Polar form is $z = \left(\cos\dfrac{3\pi}{4} + i\sin\dfrac{3\pi}{4}\right)$ and

Exponential form is $z = \sqrt{2}e^{\frac{3\pi}{4}i}$.

(ii) Let $z = \dfrac{i-1}{\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}} = \dfrac{i-1}{\frac{1}{2}+\frac{i\sqrt{3}}{2}} = \dfrac{2(i-1)}{(1+i\sqrt{3})}$

$\Rightarrow z = \dfrac{2(i-1)}{(1+i\sqrt{3})}\times\dfrac{(1-i\sqrt{3})}{(1-i\sqrt{3})} \Rightarrow z = \left(\dfrac{\sqrt{3}-1}{2}\right)+i\left(\dfrac{\sqrt{3}+1}{2}\right)$

$\because$ Re$(z) > 0$ and Im$(z) > 0 \Rightarrow z$ lies in first quadrant.

$\therefore |z| = \sqrt{\left(\dfrac{\sqrt{3}-1}{2}\right)^2+\left(\dfrac{\sqrt{3}+1}{2}\right)^2} = \sqrt{\dfrac{2(3+1)}{4}} = \sqrt{2}$.

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$$\tan\theta = \left|\frac{\sqrt{3}+1}{\sqrt{3}-1}\right| = \tan\frac{5\pi}{12} \Rightarrow \alpha = \frac{5\pi}{12}$$

Hence Polar form is $z = \sqrt{2}\left(\cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12}\right)$ and

Exponential form is $z = \sqrt{2}e^{\frac{5\pi}{12}i}$.

Illustration 19

If $x_n \cos\left(\frac{\pi}{2^n}\right) + i\sin\left(\frac{\pi}{2^n}\right)$, then find $x_1x_2x_3...\infty$

Solution:

$x_n = \cos\left(\frac{\pi}{2^n}\right) + i\sin\left(\frac{\pi}{2^n}\right) = 1 \times e^{i\frac{\pi}{2^n}}$

$$\therefore x_1x_2x_3...\infty = \lim_{n\to\infty}\left(e^{i\frac{\pi}{2^1}} \times e^{i\frac{\pi}{2^2}} \times e^{i\frac{\pi}{2^3}} \times ... \times e^{i\frac{\pi}{2^n}}\right) = \lim_{n\to\infty}\left(e^{i\left(\frac{\pi}{2}+\frac{\pi}{2^2}+...+\frac{\pi}{2^n}\right)}\right)$$

$= e^{i\pi} = \cos\pi + i\sin\pi = -1.$

Illustration 20

If $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2$, find amp $\left(\frac{z_1}{z_2}\right)$; $z_1z_2 \neq 0$. (Here amp(z) denotes the principle value of argument.)

Solution:

Method-I:

Putting $z = \frac{z_1}{z_2}$ in given equation, we get $|z+1|^2 = |z|^2 + 1 \Rightarrow (z+1)(\bar{z}+1) = z\bar{z} + 1$

$\Rightarrow z + \bar{z} = 0 \Rightarrow z$ is purely imaginary $\Rightarrow \text{amp}(z) = \pm\frac{\pi}{2}$

Method-II:

Here let $z_1 = r_1(\cos\theta_1 + i\sin\theta_1), |z_1| = r_1$

$z_2 = r_2(\cos\theta_2 + i\sin\theta_2), |z_2| = r_2$

$\therefore |(z_1 + z_2)|^2 = \left|(r_1\cos\theta_1 + r_2\cos\theta_2) + i(r_1\sin\theta_1 + r_2\sin\theta_2)\right|^2$

$= r_1^2 + r_2^2 + 2r_1r_2\cos(\theta_1 - \theta_2) = |z_1|^2 + |z_2|^2$ if $\cos(\theta_1 - \theta_2) = 0$ (or $r_1 = 0$)

$\therefore \theta_1 - \theta_2 = \pm\frac{\pi}{2}$ (or $r_1 = 0$) $\Rightarrow$ amp(z_1) – amp$(z_2) = \pm\frac{\pi}{2} \Rightarrow \text{amp}\left(\frac{z_1}{z_2}\right) = \pm\frac{\pi}{2}$.

Illustration 21

If $\left(1 + i\sqrt{3}\right)^9 = a + ib$; $a, b \in R$, then find the value of $a + 10b$.

Solution:

$$1 + i\sqrt{3} = 2\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = 2\left[\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right] = 2e^{i\pi/3}$$

$$\therefore \left(1 + i\sqrt{3}\right)^9 = \left(2e^{i\pi/3}\right)^9 = 2^9.e^{i(3\pi)} = 2^9(\cos3\pi + i\sin3\pi) = -2^9$$

$\therefore a + ib = \left(1 + i\sqrt{3}\right)^9 = -2^9 \Rightarrow a + 10b = -2^9.$

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Illustration 22

If $\cos\alpha + \cos\beta + \cos\gamma = 0$ and also $\sin\alpha + \sin\beta + \sin\gamma = 0$, then prove the following :

(a) $\cos2\alpha + \cos2\beta + \cos2\gamma = \sin2\alpha + \sin2\beta + \sin2\gamma = 0$

(b) $\sin3\alpha + \sin3\beta + \sin3\gamma = 3\sin(\alpha + \beta + \gamma)$

(c) $\cos3\alpha + \cos3\beta + \cos3\gamma = 3\cos(\alpha + \beta + \gamma)$

Solution:

Let $z_1 = \cos\alpha + i\sin\alpha$, $z_2 = \cos\beta + i\sin\beta$ & $z_3 = \cos\gamma + i\sin\gamma$.

$\therefore z_1 + z_2 + z_3 = (\cos\alpha + \cos\beta + \cos\gamma) + i(\sin\alpha + \sin\beta + \sin\gamma)$

$= 0 + i \cdot 0 = 0$ ...(i)

(a) Also $\frac{1}{z_1} = (\cos\alpha + i\sin\alpha)^{-1} = \cos\alpha - i\sin\alpha$

$\frac{1}{z_2} = \cos\beta - i\sin\beta$, $\frac{1}{z_3} = \cos\gamma - i\sin\gamma$

$\therefore \frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} = (\cos\alpha + \cos\beta + \cos\gamma) - i(\sin\alpha + \sin\beta + \sin\gamma)$ ...(ii)

$= 0 - i \cdot 0 = 0$

Now $z_1^2 + z_2^2 + z_3^2 = (z_1 + z_2 + z_3)^2 - 2(z_1z_2 + z_2z_3 + z_3z_1)$

$= 0 - 2z_1z_2z_3\left(\frac{1}{z_3} + \frac{1}{z_1} + \frac{1}{z_2}\right) = 0 - 2z_1z_2z_3 \cdot 0 = 0$ {using (i) and (ii)}

or $(\cos\alpha + i\sin\alpha)^2 + (\cos\beta + i\sin\beta)^2 + (\cos\gamma + i\sin\gamma)^2 = 0$

or $\cos2\alpha + i\sin2\alpha + \cos2\beta + i\sin2\beta + \cos2\gamma + i\sin2\gamma = 0 + i.0$

Equating real and imaginary parts on both sides,

$\cos2\alpha + \cos2\beta + \cos2\gamma = 0$ and $\sin2\alpha + \sin2\beta + \sin2\gamma = 0$

(b) If $z_1 + z_2 + z_3 = 0$ then $z_1^3 + z_2^3 + z_3^3 = 3z_1z_2z_3$

$\therefore (\cos\alpha + i\sin\alpha)^3 + (\cos\beta + i\sin\beta)^3 + (\cos\gamma + i\sin\gamma)^3$

$= 3(\cos\alpha + i\sin\alpha)(\cos\beta + i\sin\beta)(\cos\gamma + i\sin\gamma)$

or $\cos3\alpha + i\sin3\alpha + \cos3\beta + i\sin3\beta + \cos3\gamma + i\sin3\gamma$

$= 3\{\cos(\alpha + \beta + \gamma) + i\sin(\alpha + \beta + \gamma)\}$

Equating imaginary parts on both sides, $\sin3\alpha + \sin3\beta + \sin3\gamma = 3\sin(\alpha + \beta + \gamma)$

(c) Equating real parts on both sides, $\cos3\alpha + \cos3\beta + \cos3\gamma = 3\cos(\alpha + \beta + \gamma)$

8. GEOMETRICAL APPLICATIONS OF MODULUS:

8.1 Geometrical Meaning of Modulus:

(i) The modulus of a complex number z represents distance of point $P(z)$ from origin i.e. If $z = x + iy; x, y \in \mathbb{R}$, then $|z| = \sqrt{x^2 + y^2} = OP$, where $O(0,0), P(x,y)$.

(ii) If z_1 and z_2 are complex numbers, then $|z_1 - z_2|$ represents distance between points $P(z_1)$ and $Q(z_2)$ i.e. If $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, then $|z_1 - z_2| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} = PQ$, where $P(x_1, y_1)$ and $Q(x_2, y_2)$.

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Illustration 23

Plot locus of z satisfying following equations

(a) $|z| = 2$

(b) $|z - 1 - 2i| = 3$

(c) $|z - 2i| = |z + 3|$

(d) $|z - 4| + |z + 3i| = 5$

(e) $|z - 3i| - |z + 4| = 5$

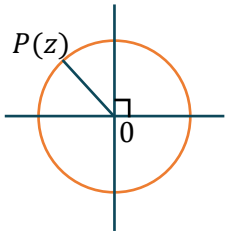
Solution:

(a) $|z| = 2$

$\Rightarrow$ Distance of point $P(z)$ from origin $= 2$

$\Rightarrow$ Point $P(z)$ moves such that it is always at 2 units from origin.

$\Rightarrow$ Locus of z is a circle with centre at (0,0) and radius 2.



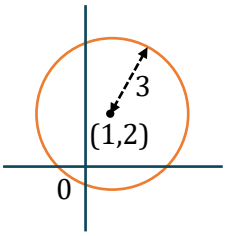
(b) $|z - 1 - 2i| = 3$

$\Rightarrow$ Distance of point $P(z)$ from $A(1 + 2i) = 3$

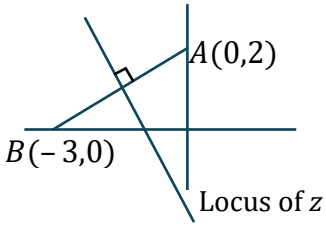
$\Rightarrow$ Point $P(z)$ moves such that it is always at 3 units from $A(1, 2)$.

$\Rightarrow$ Locus of z is a circle with centre $A(1,2)$ and radius 3.

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(c) $|z - 2i| = |z + 3|$



$\Rightarrow$ Distance of $P(z)$ from $A(2i)$

= Distance of $P(z)$ from $B(-3,0)$

$\Rightarrow$ $P(z)$ is equidistant from points $A(0,2)$ and $B(-3,0)$

$\Rightarrow$ P lies on perpendicular bisector of line segment AB.

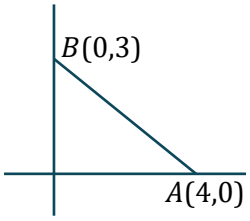
i.e. line $6x + 4y + 5 = 0$

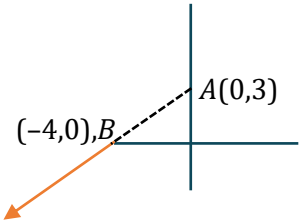
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(d) $|z-4|+|z-3i|=5$ is of form $PA+PB=AB$.

where $P(z)$, $A(4,0)$ and $B(0,3)$.

Thus P can be anywhere on the line segment AB.





Hence locus of z is line segment AB

(e) Given equation is of form $PA-PB=AB$ where

$P(z)$, $A(0,3)$, $B(-4,0)$.

Hence locus of z is a ray starting from A and in direction of $\overrightarrow{AB}$

8.2 Locus Involving Modulus:

Let affixes of points P, A, B are z, z_1, z_2. Here P is a variable point and A, B are fixed points. Also let a, k, r, r_1, r_2 are positive constants. Then observe the following points.

(i) CIRCLE:

Locus of $P(z)$ satisfying $|z|=r$ is a circle with centre at (0,0) and radius r. Based on this, we can conclude the following results:

(a) Locus of $P(z)$ satisfying $|z-z_1|=r$ is a circle with center $A(z_1)$ and radius r.

(b) Locus of $P(z)$ satisfying $|z-z_1|<r$ is set of all the points inside the circle with center $A(z_1)$ and radius r,

(c) Locus of $P(z)$ satisfying $|z-z_1|>r$ is set of all the points outside the circle with center $A(z_1)$ and radius r

(d) Locus of $P(z)$ satisfying $r_1<|z-z_1|<r_2$ represents the region of argand plane lying between the two concentric circles $|z-z_1|=r_1$, $|z-z_1|=r_2$

(e) The equation of circle with centre at z_0 and radius r is $|z-z_0|=r$

$\Rightarrow (z-z_0)(\bar{z}-\bar{z}_0)=r^2 \Rightarrow z\bar{z}-z_0\bar{z}-\bar{z}_0z+z_0\bar{z}_0-r^2=0$,

(f) General equation of circle is $z\bar{z}+\bar{\alpha}z+\alpha\bar{z}+\beta=0$. ($\alpha, z \in \mathbb{C}$, $\beta \in \mathbb{R}$) with center '$-\alpha$', radius $\sqrt{\alpha\bar{\alpha}-\beta}$

(ii) ELLIPSE:

Locus of z satisfying $|z-z_1|+|z-z_2|=2a$ when $2a>|z_1-z_2|$ is an ellipse with

(a) z_1, z_2 as foci,

(b) '$2a$' as length of major axis

(c) eccentricity $e=\dfrac{|z_1-z_2|}{2a}$.

(d) When $2a=|z_1-z_2|$, Locus of z is the line segment joining z_1, z_2.

(e) There cannot be any point in the argand plane such that $2a<|z_1-z_2|$

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(iii) HYPERBOLA:

Locus of z satisfying $||z - z_1| - |z - z_2|| = 2a$, when $2a < |z_1 - z_2|$, is a hyperbola with

(a) z_1, z_2 as foci,

(b) '$2a$' as length of transverse axis

(c) eccentricity $e = \dfrac{|z_1 - z_2|}{2a}$.

(d) In case, $|z_1 - z_2| = 2a$, locus of z is the line joining z_1 and z_2 excluding the line segment joining z_1 and z_2 (points z_1, z_2 are included).

(e) There cannot be any point in the argand plane such that $2a > |z_1 - z_2|$

(iv) PERPENDICULAR BISECTOR :

Locus of $P(z)$ satisfying $\left|\dfrac{z - z_1}{z - z_2}\right| = 1$ i.e. $|z - z_1| = |z - z_2|$ i.e. $PA = PB$ is the perpendicular bisector of the line segment joining $A(z_1)$ and $B(z_2)$.

(v) APOLLONIUS CIRCLE:

Locus of $P(z)$ satisfying $\left|\dfrac{z - z_1}{z - z_2}\right| = k$ i.e. $\dfrac{PA}{PB} = k$ $(k \neq 1, > 0)$ a circle called Apollonius Circle, whose center divides AB in ratio $k^2 : 1$ externally and radius $\left|\dfrac{k \cdot AB}{k^2 - 1}\right|$.

(vi) DIAMETRICAL FORM OF CIRCLE:

Locus of $P(z)$ satisfying $|z - z_1|^2 + |z - z_2|^2 = |z_1 - z_2|^2$ i.e. $PA^2 + PB^2 = AB^2$ is a circle with A, B as ends of diameter.

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Illustration 24

The locus of the complex number z in argand plane satisfying the inequality

$$\log_{\frac{1}{2}}\left(\frac{|z-1|+4}{3|z-1|-2}\right) > 1 \quad \left(\text{where } |z-1| \neq \frac{2}{3}\right) \text{ is}$$

(A) a circle (B) interior of a circle

(C) exterior of a circle (D) part of a line

Solution:

We have, $\log_{1/2}\left(\dfrac{|z-1|+4}{3|z-1|-2}\right) > 1 = \log_{1/2}\left(\dfrac{1}{2}\right)$

$\Rightarrow \dfrac{|z-1|+4}{3|z-1|-2} < \dfrac{1}{2}$ [$\because \log_a x$ is a decreasing function if $0 < a < 1$]

$\Rightarrow 2|z - 1| + 8 < 3|z - 1| - 2$ as $|z - 1| > 2/3$

$\Rightarrow |z - 1| > 10$ which is exterior of a circle with center 1 and radius 10.

Illustration 25

Find locus of z satisfying

(a) $|z-2-3i| > 1$ (b) $|z-2i+3| < 2$ (c) $2 \le |z-2i| < 3$

(d) $\left|\dfrac{z-i}{z+i}\right| = 1$ (e) $\left|\dfrac{z-i}{z+2}\right| = 2$ (f) $|z-4i|^2 + |z+3|^2 = 25$

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Solution:

(a) Let $P(z), A(2,3)$. Then $|z-2-3i|>1$ means $PA>1$ i.e. distance of point P from A is always greater than 1 i.e. P lies outside a circle with center A and radius 1.

(b) Let $P(z), A(-3,2)$. Then $|z-2i+3|<2$ means $PA<2$ i.e. distance of point P from A is always less than 2 i.e. P lies inside a circle with center A and radius 2.

(c) Let $P(z), A(0,2)$. Then $2\le|z-2i|<3$ means $2\le PA<3$ i.e. distance of point P from A lies in the interval [2, 3) i.e. P lies on or outside the circle with center A, radius 2, but inside the circle with center A, radius 3.

(d) Putting $z=x+iy; x,y\in\mathbb{R}$, we get $\left|\frac{z-i}{z+i}\right|=1\Rightarrow\left|\frac{x+i(y-1)}{x+i(y+1)}\right|=1$

$$\Rightarrow\frac{|x+i(y-1)|^2}{|x+i(y+1)|^2}=1\Rightarrow x^2+(y-1)^2=x^2+(y+1)^2\Rightarrow 4y=0\Rightarrow y=0$$

Hence locus of z is Real Axis, also the perpendicular bisector of segment joining (0, 1) and (0, -1).

(e) Putting $z=x+iy; x,y\in\mathbb{R}$, we get $\left|\frac{z-i}{z+2}\right|=2\Rightarrow\left|\frac{x+i(y-1)}{(x+2)+iy}\right|=2$

$$\Rightarrow x^2+(y-1)^2=4\left((x+2)^2+y^2\right)\Rightarrow x^2+y^2+\frac{16}{3}x+\frac{2}{3}y+5=0$$

Hence locus of z is a circle with center $\left(-\frac{8}{3},-\frac{1}{3}\right)$ and radius $\frac{\sqrt{20}}{3}$.

(f) Putting $z=x+iy; x,y\in\mathbb{R}$, we get $|z-4i|^2+|z+3|^2=25$

$$\Rightarrow x^2+(y-4)^2+(x+3)^2+y^2=25\Rightarrow x^2+y^2+3x-4y=0$$

Also note that the given equation is of form $PA^2+PB^2=AB^2$.

Hence locus of z is a circle with $A(4i)$ and $B(-3, 0)$ as ends of diameter.

Illustration 26

Find locus of z satisfying

(a) $|z-1|+|z-2i|=\sqrt{5}$ (b) $|z-1+i|+|z+2-3i|=10$

(c) $|z-i|+|z+2|=2$ (d) $\big||z+i|-|z+2|\big|=\sqrt{5}$

(e) $\big||z+i-1|-|z+2-3i|\big|=4$ (f) $\big||z+2i-1|-|z+2-2i|\big|=7$

Solution:

(a) Let $P(z), A(1,0), B(0,2)$. Then $|z-1|+|z-2i|=\sqrt{5}$ means $PA+PB=AB$.

Hence locus of z will be the line segment AB.

(b) Let $P(z), A(1,-1), B(-2,3)$. Then $|z-1+i|+|z+2-3i|=10$ means $PA+PB=10$, where $AB=5$.

Hence locus of z will be an ellipse with eccentricity $\frac{1}{2}$ and focii at A, B.

(c) Let $P(z), A(0,1), B(-2,0)$. Then $|z-i|+|z+2|=2$ means

$PA+PB=2$, where $AB=\sqrt{5}$.

Hence $PA+PB<AB$, which is NOT possible. Thus there exists NO point z satisfying $|z-i|+|z+2|=2$

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(d) Let $P(z), A(0,-1), B(-2,0)$. Then $\big||z+i|-|z+2|\big|=\sqrt{5}$ means $|PA-PB|=AB$.

Hence locus of z will be the line AB, excluding the points lying between A and B.

(e) Let $P(z), A(1,-1), B(-2,3)$. Then $\big||z+i-1|-|z+2-3i|\big|=4$ mean $|PA-PB|=2$, where $AB = 5$.

Hence locus of z will be a hyperbola with eccentricity $\frac{5}{2}$ and focii at A, B.

(f) Let $P(z), A(1,-2), B(-2,2)$. Then $\big||z+2i-1|-|z+2-2i|\big|=7$ means $|PA-PB|=7$, where $AB = 5$.

Hence $|PA-PB| > AB$, which is NOT possible. Thus there exists NO point z satisfying

$\big||z+2i-1|-|z+2-2i|\big|=7$.

Illustration 27

Find the complex numbers z which simultaneously satisfy the equations $\left|\frac{z-12}{z-8i}\right|=\frac{5}{3}, \left|\frac{z-4}{z-8}\right|=1$.

Solution:

Here $\left|\frac{z-4}{z-8}\right|=1 \Rightarrow \left|\frac{x-4+iy}{x-8+iy}\right|=1 \Rightarrow (x-4)^2+y^2=(x-8)^2+y^2 \Rightarrow x=6$

With $x = 6$, $\left|\frac{z-12}{z-8i}\right|=\frac{5}{3}$

$\Rightarrow \left|\frac{-6+iy}{6+i(y-8)}\right|=\frac{5}{3} \Rightarrow 9(36+y^2)=25[36+(y-8)^2] \Rightarrow y^2-25y+136=0$

$\Rightarrow y = 17, 8$

Hence the required numbers are $z = 6 + 17i, 6 + 8i$

Illustration 28

If $z = x + iy$ and $\omega=\frac{1-iz}{z-i}$, (where $i=\sqrt{-1}$) then prove that $|\omega| = 1$ implies that in the complex plane z lies on real axis.

Solution:

Given, $\omega=\frac{1-iz}{z-i} \Rightarrow |\omega|=\left|\frac{1-iz}{z-i}\right| \Rightarrow 1=\left|\frac{1-iz}{z-i}\right| \Rightarrow |z-i|^2=|1-iz|^2$

$\Rightarrow (z-i)(\bar{z}+i)=(1-iz)(1+i\bar{z}) \Rightarrow z\bar{z}+iz-i\bar{z}+1=1+i\bar{z}-iz+z\bar{z}$

$\Rightarrow 2iz-2i\bar{z}=0 \Rightarrow z=\bar{z}$

Hence, z lies on real axis

Illustration 29

Let z_1, z_2, z_3 be three distinct complex numbers satisfying $|z_1 - 1| = |z_2 - 1| = |z_3 - 1|$. Let A, B and C be the points represented in the Argand plane corresponding to z_1, z_2 and z_3 respectively. Prove that $z_1 + z_2 + z_3 = 3$ if and only if ΔABC is an equilateral triangle.

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Solution:

$|z_1 - 1| = |z_2 - 1| = |z_3 - 1|$

$\Rightarrow$ The point corresponding to 1 (say P) is equidistant from the points A, B and C.

$\Rightarrow$ $P(1,0)$ is the circumcentre of the ΔABC

Now if $z_1 + z_2 + z_3 = 3$ then the point corresponding to centroid of the ΔABC is $\frac{z_1 + z_2 + z_3}{3} = 1$.

$\Rightarrow$ circumcentre and centroid coincide.

$\Rightarrow$ ΔABC is equilateral

Illustration 30

Let a be a complex number such that $|a| < 1$ and $z_1, z_2, \ldots\ldots, z_n$ be the vertices of a polygon such that $z_k = 1 + a + a^2 + \ldots. a^k$, then show that vertices of the polygon lie within the circle $\left|z - \frac{1}{1-a}\right| = \frac{1}{|1-a|}$.

Solution:

We have $z_k = 1 + a + a^2 + \ldots\ldots + a^k = \frac{1 - a^{k+1}}{1 - a}$

$\Rightarrow \quad z_k - \frac{1}{1-a} = \frac{-a^{k+1}}{1-a} \Rightarrow \left|z_k - \frac{1}{1-a}\right| = \frac{|a|^{k+1}}{|1-a|} < \frac{1}{|1-a|} \qquad (\because |a| < 1)$

$\therefore$ Vertices of the polygon $z_1, z_2, \ldots\ldots, z_n$ lie within the circle $\left|z - \frac{1}{1-a}\right| = \frac{1}{|1-a|}$

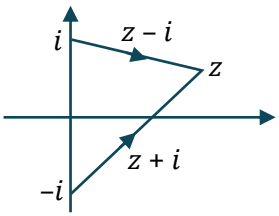
Illustration 31

If $\alpha = \frac{z - i}{z + i}$, show that, when z lies above the real axis, α will lie within the unit circle which has centre at the origin. Find the locus of α if z lies on the real axis.

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Solution:

From figure, it is clear that $|z - i| < |z + i|$ (as z lies above the real axis).



$\Rightarrow |\alpha| = \frac{|z - i|}{|z + i|} < 1 \Rightarrow |\alpha| < 1$

$\therefore$ α lies within the unit circle which has centre at the origin.

Now if z lies on real axis, then $\text{Im}(z) = 0$

Let $z = x + i\,0 \Rightarrow \alpha = \frac{x - i}{x + i}$

$\Rightarrow |\alpha| = \frac{|x - i|}{|x + i|} = 1$ (as $|x - 1| = |x + i|\ \forall\ x \in R$).

$\Rightarrow$ α moves on the unit circle which has centre at the origin.

8.3 Triangle's Inequality

For any two complex numbers z_1 and z_2, the following result always holds :

$||z_1| - |z_2|| \le |z_1 + z_2| \le |z_1| + |z_2|$ and $||z_1| - |z_2|| \le |z_1 - z_2| < |z_1| + |z_2|$.

Illustration 32

If $|z| = 10$, then find the range of possible values of

(a) $|z - 2 - 3i|$ (b) $|iz + 4 - 3i|$ (c) $\left|\frac{i - z}{12 - 5i}\right|$

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Solution:

Using $||z_1| - |z_2|| \le |z_1 + z_2| \le |z_1| + |z_2|$, we get

(a) $\big||z|-|2+3i|\big| \le |z-2-3i| \le |z|+|2+3i| \Rightarrow 10-\sqrt{13} \le |z-2-3i| \le 10+\sqrt{13}$

(b) $|iz+4-3i| = |iz-4i^2-3i| = |z-4i-3|\cdot|i| = |z-4i-3|$

$\Rightarrow \big||z|-|4i+3|\big| \le |z-4i-3| \le |z|+|4i+3|$

$\Rightarrow 5 \le |z-4i-3| \le 15$

(c) $\left|\dfrac{i-z}{12-5i}\right| = \dfrac{|i-z|}{|12-5i|} = \dfrac{1}{13}|z-i|$

Also, $\big||z|-|i|\big| \le |z-i| \le |z|+|i| \Rightarrow 9 \le |z-i| \le 11 \Rightarrow \dfrac{9}{13} \le \left|\dfrac{i-z}{12-5i}\right| \le \dfrac{11}{13}$

Illustration 33

If $|z-3-4i| = 10$, then find the range of possible values of

(a) $|z-2i-3|$ (b) $|iz+4i-3|$ (c) $\left|\dfrac{1-iz}{4i-3}\right|$.

Solution:

Let $z-3-4i = w$. Then | w | = 10

(a) $|z-2i-3|$ $|z-2i-3| = |w+2i|$

Then $\big||w|-|2i|\big| \le |w+2i| \le |w|+|2i| \Rightarrow 8 \le |w+2i| \le 12$

$\Rightarrow |z-2-3i| \in [8,12]$

(b) $|iz+4i-3| = |iz+4i+3i^2| = |z+4+3i| = |w+7+7i|$

Now, $\big||w|-|7+7i|\big| \le |w+7+7i| \le |w|+|7+7i|$

$\Rightarrow 10-7\sqrt{2} \le |w+7+7i| \le 10+7\sqrt{2}$

$\Rightarrow 10-7\sqrt{2} \le |iz+4i-3| \le 10+7\sqrt{2}$

(c) $\left|\dfrac{1-iz}{4i-3}\right| = \dfrac{|1-iz|}{|4i-3|} = \dfrac{|z+i|}{5} = \dfrac{|w+3+5i|}{5}$

Now using $\big||w|-|3+5i|\big| \le |w+3+5i| \le |w|+|3+5i|$, we get $5 \le |w+3+5i| \le 15$

$\Rightarrow \left|\dfrac{1-iz}{4i-3}\right| \in [1,3]$

Illustration 34

If $\left|z-\dfrac{4}{z}\right| = 2$, then find the greatest value of $|z|$

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Solution:

We have $|z| = \left|z - \frac{4}{z} + \frac{4}{z}\right| \le \left|z - \frac{4}{z}\right| + \frac{4}{|z|} = 2 + \frac{4}{|z|}$

$\Rightarrow |z|^2 \le 2|z| + 4 \Rightarrow (|z| - 1)^2 \le 5$

$\Rightarrow |z| - 1 \le \sqrt{5} \Rightarrow |z| \le \sqrt{5} + 1$

Therefore, the greatest value of $|z|$ is $\sqrt{5}$ + 1.

Illustration 35

If z_1 and z_2 are two complex numbers and $C > 0$, then prove that

$|z_1 + z_2|^2 \le (1 + C)\,|z_1|^2 + (1 + C^{-1})|z_2|^2$.

Solution:

We have to prove that : $|z_1 + z_2|^2 \le (1 + C)\,|z_1|^2 + (1 + C^{-1})|z_2|^2$

i.e. $|z_1|^2 + |z_2|^2 + z_1\bar{z}_2 + \bar{z}_1 z_2 \le (1 + C)\,|z_1|^2 + (1 + C^{-1})|z_2|^2$

or $z_1\bar{z}_2 + \bar{z}_1 z_2 \le C|z_1|^2 + C^{-1}|z_2|^2$

or $C|z_1|^2 + \frac{1}{C}|z_2|^2 - z_1\bar{z}_2 - \bar{z}_1 z_2 \ge 0$ (using Re $(z_1\bar{z}_2) \le |z_1\bar{z}_2|$)

or $\left(\sqrt{C}|z_1| - \frac{1}{\sqrt{C}}|z_2|\right)^2 \ge 0$ which is always true.

Illustration 36

If $\theta \in \left[\frac{\pi}{6}, \frac{\pi}{3}\right]$, i = 1, 2, 3, 4, 5 and $z^4\cos\theta_1 + z^3\cos\theta_2 + z^2\cos\theta_3 + z\cos\theta_4 + \cos\theta_5 = 2\sqrt{3}$, then show that $|z| > \frac{3}{4}$.

Solution:

Given that $\cos\theta_1 . z^4 + \cos\theta_2 . z^3 + \cos\theta_3 . z^2 + \cos\theta_4 . z + \cos\theta_5 = 2\sqrt{3}$

or $\left|\cos\theta_1 . z^4 + \cos\theta_2 . z^3 + \cos\theta_3 . z^2 + \cos\theta_4 . z + \cos\theta_5\right| = 2\sqrt{3}$]

$2\sqrt{3} \le \left|\cos\theta_1 . z^4\right| + \left|\cos\theta_2 . z^3\right| + \left|\cos\theta_3 . z^2\right| + \left|\cos\theta_4 . z\right| + \left|\cos\theta_5\right|$

$\because\ \theta_i \in [\pi/6, \pi/3]$

$\therefore\ \frac{1}{2} \le \cos\theta_i \le \frac{\sqrt{3}}{2}$

$2\sqrt{3} \le \frac{\sqrt{3}}{2}|z|^4 + \frac{\sqrt{3}}{2}|z|^3 + \frac{\sqrt{3}}{2}|z|^2 + \frac{\sqrt{3}}{2}|z| + \frac{\sqrt{3}}{2}$

$\Rightarrow 3 \le |z|^4 + |z|^3 + |z|^2 + |z| \Rightarrow 3 < |z| + |z|^2 + |z|^3 + |z|^4 + |z|^5 + \ldots\ldots\infty$

$\Rightarrow 3 < \frac{|z|}{1 - |z|} \Rightarrow 3 - 3|z| < |z| \Rightarrow 4|z| > 3$

$\therefore\ |z| > \frac{3}{4}$

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9. COMPLEX NUMBER AS A POINT:

9.1 Section Formula:

Let $P(z_1)$ and $Q(z_2)$ be two given points. Let $R(z)$ be the point which divides the join of P and Q in ratio $\lambda : 1$.

- If R divides PQ internally in $\lambda : 1$, then $z = \dfrac{z_1 + \lambda z_2}{1+\lambda}$ $(\lambda + 1 \neq 0)$
- If R divides PQ externally in $\lambda : 1$, then $z = \dfrac{z_1 - \lambda z_2}{1-\lambda}$ $(\lambda \neq 1)$
- If R is midpoint of PQ, then $z = \dfrac{z_1 + z_2}{2}$
- If ℓ, m, n are three real numbers such that $\ell z_1 + m z_2 + n z_3 = 0$ and $\ell + m + n = 0$, where ℓ, m, n are not all zeros simultaneously, then the points z_1, z_2, z_3 are collinear.

9.2 Area of Triangle:

Let z_1, z_2, z_3 be vertices of a ΔABC, then area of $\Delta ABC = \left| \dfrac{i}{4} \begin{vmatrix} z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \\ z_3 & \bar{z}_3 & 1 \end{vmatrix} \right|$

9.3 Special Points of Triangle:

Let $A(z_1), B(z_2)$ and $C(z_3)$ be vertices of ΔABC. Then

(a) Centroid of $\Delta ABC = \dfrac{z_1 + z_2 + z_3}{3}$

(b) Orthocentre of the ΔABC is $\dfrac{z_1 \tan A + z_2 \tan B + z_3 \tan C}{\tan A + \tan B + \tan C}$ whenever defined.

(c) Incentre of the ΔABC is $\dfrac{az_1 + bz_2 + cz_3}{a+b+c}$, where $a = BC, b = CA, c = AB$.

(d) Circumcentre of the ΔABC is $\dfrac{z_1 \sin 2A + z_2 \sin 2B + z_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C}$

9.4 Parallelogram:

If z_1, z_2, z_3, z_4 are the affixes of the vertices A, B, C, D respectively of parallelogram $ABCD$, then $z_1 + z_3 = z_2 + z_4$

Illustration 37

Prove that the complex number $\dfrac{z_1 + z_2 + z_3}{3}$ represents the centroid of triangle formed by points represented by complex numbers z_1, z_2 and z_3.

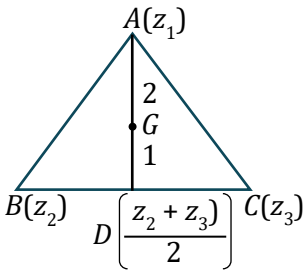
Solution:

Let G be the centroid and let the line joining A and G meet the line BC at the point D.

We have, $BD = DC$, $D \equiv \dfrac{z_2 + z_3}{2}$

G divides AD internally in ratio $2 : 1$

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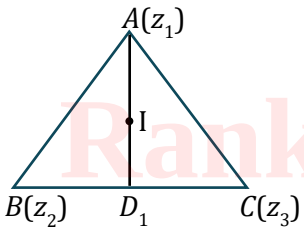
$$\Rightarrow G \equiv \frac{\frac{2(z_2+z_3)}{2}+z_1}{2+1} \equiv \frac{z_1+z_2+z_3}{3}$$

Illustration 38

Prove that the complex number $\frac{az_1+bz_2+cz_3}{a+b+c}$ represents the incentre of triangle formed by points represented by complex numbers z_1, z_2 and z_3,where $a = BC, b = CA, c = AB$.

Solution:

Let I be the incentre and let the line connecting A and I meet the line BC at D_1. We have already studied that $\frac{BD_1}{D_1C} = \frac{AB}{AC} = \frac{c}{b}$ and $\frac{AI}{ID_1} = \frac{AB+AC}{BC} = \frac{c+b}{a}$.



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Hence, by using section formula, we get

$$\Rightarrow D_1 = \frac{bz_2+cz_3}{b+c} \text{ and } I = \frac{az_1+bz_2+cz_3}{a+b+c}$$

Illustration 39

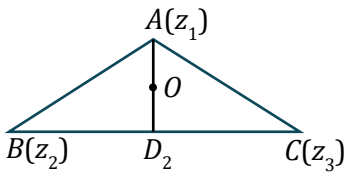
Prove that the complex number $\frac{z_1\sin 2A + z_2\sin 2B + z_3\sin 2C}{\sin 2A+\sin 2B+\sin 2C}$ represents the circumcenter of triangle formed by points represented by complex numbers z_1, z_2 and z_3.

Solution:

Let 'O' be the circumcentre and let the line connecting A and O meet the line BC at D_2.

We know that $\frac{BD_2}{D_2C} = \frac{\sin 2C}{\sin 2B}$

and $\frac{AO}{OD_2} = \frac{\sin 2B+\sin 2C}{\sin 2A}$



Hence, by using Section formula, we get $D_2 = \frac{z_2\sin 2B + z_3\sin 2C}{\sin 2B+\sin 2C}$

and $O = \frac{D_2(\sin 2B+\sin 2C)+z_1\sin 2A}{\sin 2A+\sin 2B+\sin 2C} = \frac{z_1\sin 2A+z_2\sin 2B+z_3\sin 2C}{\sin 2A+\sin 2B+\sin 2C}$

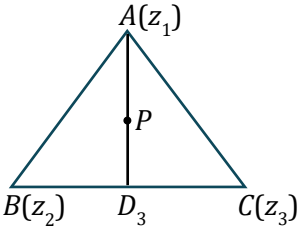
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Illustration 40

Prove that the complex number $\frac{z_1 \tan A + z_2 \tan B + z_3 \tan C}{\tan A + \tan B + \tan C}$ represents the orthocenter of a non-right triangle formed by points represented by complex numbers z_1, z_2 and z_3.

Solution:

Let 'P' be the orthocentre and let the line connecting the points A and P meet the line BC and D_3.



We have, $\frac{BD_3}{D_3C} = \frac{\tan C}{\tan B}$

and $\frac{AP}{PD_3} = \frac{\tan C + \tan B}{\tan A}$

$\Rightarrow D_3 = \frac{z_2 \tan B + z_3 \tan C}{\tan B + \tan C}$

and $P = \frac{D_3(\tan B + \tan C) + z_1 \tan A}{\tan A + \tan B + \tan C} = \frac{z_1 \tan A + z_2 \tan B + z_3 \tan C}{\tan A + \tan B + \tan C}$

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9.5 General Locus Problems

Observe the following Illustrations:

Illustration 41

Find the locus of z satisfying $Re(z^2) = 0$

Solution:

Let $z = x + iy$

$\Rightarrow z^2 = x^2 - y^2 + 2xyi$

$\therefore Re(z^2) = 0$

$\Rightarrow x^2 - y^2 = 0 \Rightarrow y = \pm x$

Thus $Re(z^2) = 0$ represents a pair of straight lines passing through origin.

Illustration 42:

If z is a complex number such that $z^2 = (\bar{z})^2$, then find locus of z.

Solution:

Let $z = x + iy$, then its conjugate $\bar{z} = x - iy$

Given that $z^2 = (\bar{z})^2 \Rightarrow x^2 - y^2 + 2ixy = x^2 - y^2 - 2ixy \Rightarrow 4ixy = 0$

Hence, locus of z is union of real and imaginary axis.

Illustration 43

Find all non-zero complex numbers satisfying $\bar{Z} = iZ^2$.

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Solution:

Let $Z = x + iy$; $\bar{Z} = x - iy$; $Z^2 = x^2 - y^2 + 2ixy$

$\therefore$ The equation is $x - iy = i(x^2 - y^2 + 2ixy)$

Equating real and imaginary parts, we get

$x = -2xy$ ...(i)

$-y = x^2 - y^2$ ...(ii)

(i) gives either $x = 0$, in that case $y = 0$; $y = 1$

or $y = -\frac{1}{2}$, in that case $\frac{1}{4} + \frac{1}{2} = x^2$

$\therefore\ x = \pm\frac{\sqrt{3}}{2}$

The non-zero Z, satisfying the equation are

$Z_1 = i$; $Z_2 = \frac{\sqrt{3}}{2} - \frac{1}{2}i$; $Z_3 = -\frac{\sqrt{3}}{2} - \frac{1}{2}i$

Illustration 44

Find the locus of z satisfying $|z - 1|^2 + |z + 1|^2 = 4$.

Solution:

Let $z = x + iy$

$\Rightarrow |x + iy - 1|^2 + |x + iy + 1|^2 = 4$

$\Rightarrow (x - 1)^2 + y^2 + (x + 1)^2 + y^2 = 4$

$\Rightarrow x^2 - 2x + 1 + y^2 + x^2 + 2x + 1 + y^2 = 4 \Rightarrow x^2 + y^2 = 1$

Hence, locus of z is circle with center at origin and radius unity.

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Illustration 45

Find all complex numbers z which satisfy $|z|^2 - 2iz + 2c(1 + i) = 0$;
where c is real. Also find range of c, when z has no locus.

Solution:

Put $z = a + ib$ then $a^2 + b^2 - 2ai + 2b + 2c + 2ci = 0$

$\Rightarrow (a^2 + b^2 + 2b + 2c) + (2c - 2a)i = 0$

$\Rightarrow a^2 + b^2 + 2b + 2c = 0$ and $2c - 2a = 0 \Rightarrow a = c$.

Now $b^2 + 2b + (c^2 + 2c) = 0 \Rightarrow\ b = -1 \pm \sqrt{1 - 2c - c^2}$

Since b is real, $1 - 2c - c^2 \geq 0 \Rightarrow\ c \in \left[-1 - \sqrt{2}, -1 + \sqrt{2}\right]$.

$\Rightarrow\ z = c + i\left(-1 \pm \sqrt{1 - 2c - c^2}\right)$

Solution of the given equation does not exits for $c \notin \left[-1 - \sqrt{2}, -1 + \sqrt{2}\right]$.

Illustration 46

Find the area bounded by the points z which satisfy $7z^2 - 50z\bar{z} + 7\bar{z}^2 + 576 = 0$.

Solution:

Let $z = x + iy$

$\Rightarrow 7\left(z^2 + \bar{z}^2\right) - 50|z|^2 + 576 = 0$

$\Rightarrow 7(2x^2 - 2y^2) - 50(x^2 + y^2) + 576 = 0$

$\Rightarrow 36x^2 + 64y^2 = 576 \Rightarrow \frac{x^2}{16} + \frac{y^2}{9} = 1$

which is an ellipse whose area 12 π sq. units

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Illustration 47

If $iz^3 + z^2 - z + i = 0$, then show that $|z| = 1$.

Solution:

$iz^3 + z^2 - z + i = 0$

By substituting $z = i$ in the equation, we get 0 = 0

Hence $z - i$ is a factor of $iz^3 + z^2 - z + i$

$\Rightarrow iz^2(z - i) - 1(z - i) = 0$

$\Rightarrow (iz^2 - 1)(z - i) = 0$

Either $iz^2 - 1 = 0$ or $z - i = 0$

When $z - i = 0, z = i \therefore |z| = |0 + i.1| = \sqrt{0^2 + 1^2} = 1$.

When $iz^2 - 1 = 0, z^2 = \frac{1}{i} = -i$

$\therefore |z^2| = |0 - l.i| = \sqrt{0^2 + (-1)^2} = 1$

$\therefore |z^2| = 1$ or $|z| = 1$

$\therefore$ In any case we have $|z| = 1$

Illustration 48

Find range of a ($\in R$) if atleast one complex number z satisfies

$|z + 3| = a^2 - 2a + 6$ and $|z - 3\sqrt{3}i| < a^2$ simultaneously.

Solution:

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$|z + 3| = a^2 - 2a + 1$

$\Rightarrow z$ lies on a circle whose centre is $(-3 + i.0)$ and radius is $a^2 - 2a + 6$

Here $a^2 - 2a + 6 > 0, \forall\ a \in R$ (as the roots of the corresponding equation are imaginary)

And $|z - 3\sqrt{3}i| < a^2$ ...(2)

$\Rightarrow z$ lies in the interior of a circle whose centre is $(0 + 3\sqrt{3}i)$ and radius is a^2.

Clearly at least one complex number would satisfy both the equations if the two circles cut in two real and distinct points or first circle lies entirely inside the second circle.

We know that two circles with centres at c_1 and c_2 and radii r_1 and r_2 respectively, will never intersect and will no overlapping region if $c_1c_2 \geq r_1 + r_2$

i.e., if $|-3 - 3\sqrt{3}i| \geq a^2 + a^2 - 2a + 6$

or if $6 \geq 2a^2 - 2a + 6$

$\Rightarrow a^2 - a \leq 0$

or if $a \leq [0, 1]$ ....(3)

Now condition for the second circle to lie entirely within the first circle is that

$r_1 - r_2 \geq c_1c_2$

$\Rightarrow a^2 - 2a + 6 - a^2 \geq 6$

$\Rightarrow a \leq 0$ ....(4)

From (3) and (4) $a \in (-\infty, 1]$

Hence desired values of a is $R - (-\infty, 1] = (1, \infty)$

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10. GEOMETRICAL APPLICATIONS OF CONJUGATE:

If $z = x + iy; x, y \in \mathbb{R}$, then $x = \frac{z+\bar{z}}{2}, y = \frac{z-\bar{z}}{2i}$. Using these results any equation in variables x, y can be converted into an equation involving complex numbers. Observe the following Illustration:

Illustration 49

Find the equation of the straight line $y = mx + c$ (where m & c are real numbers) in complex form.

Solution:

Use $z = x + iy$, $\bar{z} = x - iy, x = \left(\frac{z+\bar{z}}{2}\right)$, $y = \left(\frac{z-\bar{z}}{2i}\right)$

$$y = mx + c \Rightarrow \frac{z-\bar{z}}{2i} = m\left(\frac{z+\bar{z}}{2}\right) + c \Rightarrow m(z+\bar{z}) + 2c + i(z-\bar{z}) = 0$$

Illustration 50

Find the equation of the parabola $y^2 = 4ax$ (where a is a real number) in complex form.

Solution:

Method-I:

Use $z = x + iy$, $\bar{z} = x - iy$, i.e. $x = \left(\frac{z+\bar{z}}{2}\right)$, $y = \left(\frac{z-\bar{z}}{2i}\right)$

$$y^2 = 4ax \Rightarrow \left(\frac{z-\bar{z}}{2i}\right)^2 = 4a\left(\frac{z+\bar{z}}{2}\right) \Rightarrow z^2 + \bar{z}^2 + 8a(z+\bar{z}) = 2|z|^2$$

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Method-II:

Every point z lying on parabola is equidistant from its focus and directrix. For parabola $y^2 = 4ax$, focus is at $S(a, 0)$ and directrix is $x + a = 0$. Thus equation of parabola can also be written as $\sqrt{(x-a)^2 + y^2} = |x + a|$

$$\Rightarrow |z - a| = |\mathrm{Re}(z + a)|$$

10.1 Equation of Line:

Complex equation of a line is the equation which identifies the complex number representing all the points lying on it.

(i) Two-Point Form:

Let L denote a line passing through $A(z_1)$ and $B(z_2)$. Also let $P(z)$ be any point lying on line L other than A,B. Then P must divide AB in some ratio say k : 1,

$$z = \frac{kz_2 + z_1}{k+1} \Rightarrow z = \left(\frac{k}{k+1}\right)z_1 + \left(\frac{1}{k+1}\right)z_2$$

This equation is of form $z = (1-t)z_1 + (t)z_2$; where $t \in \mathbb{R}$.

(ii) Parametric Form:

Let L denote a line passing through $A(z_1)$ and $B(z_2)$. Then equation of line L is $z = z_1 + t(z_1 - z_2)$; where 't' is parameter.

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(iii) General Form:

General Form of equation of line in coordinate geometry was $Ax+By+C=0\left(A,B,C\in\mathbb{R}\right)$.

Putting $x=\frac{z+\bar{z}}{2}, y=(-i)\frac{z-\bar{z}}{2}$, we get $A\left(\frac{z+\bar{z}}{2}\right)-iB\left(\frac{z-\bar{z}}{2}\right)+C=0$

$\Rightarrow z(A-Bi)+\bar{z}(A+Bi)+2C=0$

Thus, General form of line is $a\bar{z}+\bar{a}z+b=0$ $(a, z\in\mathbb{C}, b\in\mathbb{R})$.

(iv) Since P, A, B are collinear, the direction of vectors $\overrightarrow{AP}$ *and* $\overrightarrow{AB}$ must be same or opposite i.e. angle between them is 0 or π i.e.

$$\arg\left(\frac{z-z_1}{z_2-z_1}\right)=0 \text{ or } \pi \Rightarrow \frac{z-z_1}{z_2-z_1}=real \Rightarrow \frac{z-z_1}{z_2-z_1}=\frac{\bar{z}-\bar{z}_1}{\bar{z}_2-\bar{z}_1}$$

Thus, $\frac{z-z_1}{z_2-z_1}=\frac{\bar{z}-\bar{z}_1}{\bar{z}_2-\bar{z}_1}$ or $\begin{vmatrix} z & \bar{z} & 1 \\ z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \end{vmatrix}=0$ represents the equation of line joining A and B.

(v) Condition(s) of Collinearity:

If the points z_1, z_2, z_3 are coliinear, then

(a) $\frac{z_3-z_1}{z_2-z_1}=real$ (b) $\frac{z_3-z_1}{z_2-z_1}=\frac{\bar{z}_3-\bar{z}_1}{\bar{z}_2-\bar{z}_1}$

(c) $\sum\bar{z}_1(z_2-z_3)=0$ (d) $\begin{vmatrix} z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \\ z_3 & \bar{z}_3 & 1 \end{vmatrix}=0$

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Illustration 51

Prove that $-\frac{\mathrm{Re}(a)}{\mathrm{Im}(a)}$ is the slope of the line with complex equation as $a\bar{z}+\bar{a}z+b=0$ $(a, z\in\mathbb{C}, b\in\mathbb{R})$.

Solution:

Let equation of line be $Ax+By+C=0\left(A,B,C\in\mathbb{R}\right)$.

Putting $x=\frac{z+\bar{z}}{2}, y=(-i)\frac{z-\bar{z}}{2}$, we get $A\left(\frac{z+\bar{z}}{2}\right)-iB\left(\frac{z-\bar{z}}{2}\right)+C=0$

$\Rightarrow z(A-Bi)+\bar{z}(A+Bi)+2C=0$

Comparing with given equation, we get $\frac{a}{A+Bi}=\frac{\bar{a}}{A-Bi}=\frac{b}{2C}$.

Thus, slope of given line is $\frac{-A}{B}=-\frac{\mathrm{Re}(a)}{\mathrm{Im}(a)}$

Illustration 52

If a line passes through the points z_1 and z_2, then let us define its complex slope as $\omega=\frac{z_1-z_2}{\bar{z}_1-\bar{z}_2}$. Then prove that $-\frac{a}{\bar{a}}$ is the complex slope of line with complex equation as $a\bar{z}+\bar{a}z+b=0$ $(a, z\in\mathbb{C}, b\in\mathbb{R})$.

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Solution:

Let $a\bar{z}+\bar{a}z+b=0$ be equation of line joining $A(z_1)$ and $B(z_2)$. Then

$$\begin{cases} a\bar{z}_1+\bar{a}z_1+b=0 \\ a\bar{z}_2+\bar{a}z_2+b=0 \end{cases} \Rightarrow a(\bar{z}_1-\bar{z}_2)+\bar{a}(z_1-z_2)=0 \Rightarrow \omega=\frac{z_1-z_2}{\bar{z}_1-\bar{z}_2}=-\frac{a}{\bar{a}}$$

Illustration 53

If a line passes through the points z_1 and z_2, then let us define its complex slope as $\omega=\frac{z_1-z_2}{\bar{z}_1-\bar{z}_2}$. If ω_1 *and* ω_2 denote the complex slopes of two perpendicular lines, then prove that $\omega_1+\omega_2=0$. Also, prove that line with equation $\bar{\alpha}z+\alpha\bar{z}+\beta=0$ ($\alpha, z \in \mathbb{C}$, $\beta \in \mathbb{R}$) is always perpendicular to the line with equation $\bar{\alpha}z-\alpha\bar{z}+i\gamma=0$

(where $\gamma \in \mathbb{R}$).

Solution:

Let $a\bar{z}+\bar{a}z+b=0$ be equation of line L joining $A(z_1)$ and $B(z_2)$. Then we have already proved that

real slope of line L is $m=-\frac{\mathrm{Re}(a)}{\mathrm{Im}(a)}$ and

complex slope of line L is $\omega=-\frac{a}{\bar{a}}$.

Now, applying componendo and dividendo on $\omega=-\frac{a}{\bar{a}}$, we get

$$\frac{\omega-1}{\omega+1}=\frac{a+\bar{a}}{a-\bar{a}}=\frac{2\cdot\mathrm{Re}(a)}{2i\cdot\mathrm{Im}(a)}=im$$

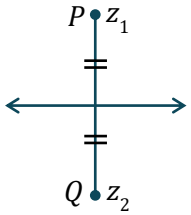
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Now, let m_1, m_2 are real slopes and ω_1, ω_2 are complex slopes of two perpendicular lines. Then $m_1m_2=-1$

$$\Rightarrow im_1\cdot im_2=1 \Rightarrow \frac{\omega_1-1}{\omega_1+1}\cdot\frac{\omega_2-1}{\omega_2+1}=1 \Rightarrow \omega_1+\omega_2=0 \ .$$

Illustration 54

Prove that the two points denoted by the complex numbers z_1 & z_2 will be the reflection of each other in the line $\bar{\alpha}z+\alpha\bar{z}+r=0$ if and only if $\bar{\alpha}z_1+\alpha\bar{z}_2+r=0$, where r is real and α is non zero complex constant.



Solution:

Proof of given statement:

$P(z_1)$ and $Q(z_2)$ are reflections of each other in the line $L:\bar{\alpha}z+\alpha\bar{z}+r=0$.

Then mid-point of line PQ lies on L

$$\Rightarrow \bar{\alpha}\left(\frac{z_1+z_2}{2}\right)+\alpha\left(\frac{\bar{z}_1+\bar{z}_2}{2}\right)+r=0 \Rightarrow (\bar{\alpha}z_1+\alpha\bar{z}_2+r)+\overline{(\bar{\alpha}z_1+\alpha\bar{z}_2+r)}=0$$

$$\Rightarrow \mathrm{Re}(\bar{\alpha}z_1+\alpha\bar{z}_2+r)=0$$

Also, $PQ\perp L \Rightarrow \omega_{PQ}+\omega_L=0$

$$\Rightarrow \frac{z_1-z_2}{\bar{z}_1-\bar{z}_2}-\frac{\alpha}{\bar{\alpha}}=0 \Rightarrow \alpha(\bar{z}_1-\bar{z}_2)=\bar{\alpha}(z_1-z_2) \Rightarrow (\bar{\alpha}z_1+\alpha\bar{z}_2)=\overline{(\bar{\alpha}z_1+\alpha\bar{z}_2)}$$

$$\Rightarrow \mathrm{Im}(\bar{\alpha}z_1+\alpha\bar{z}_2+r)=0$$

Hence proved that $\bar{\alpha}z_1+\alpha\bar{z}_2+r=0$ as both its real and imaginary parts are zero.

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Proof of Converse:

$\bar{\alpha} z + \alpha \bar{z} + r = 0$ and $\bar{\alpha} z_1 + \alpha \bar{z}_2 + r = 0$

$$\Rightarrow \bar{\alpha}(z - z_1) + \alpha(\bar{z} - \bar{z}_2) = 0 \Rightarrow \left|\frac{z - z_1}{\bar{z} - \bar{z}_2}\right| = \left|-\frac{\alpha}{\bar{\alpha}}\right| = 1$$

$\Rightarrow |z - z_1| = |\bar{z} - \bar{z}_2| \Rightarrow |z - z_1| = |z - z_2|$, which is the equation of perpendicular bisector of PQ.

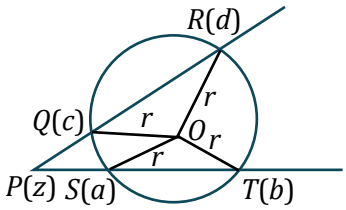
Hence proved that $P(z_1)$ and $Q(z_2)$ are reflections of each other in the line $L: \bar{\alpha} z + \alpha \bar{z} + r = 0$.

Illustration 55

Two different non-parallel lines cut the circle $|z| = r$ in points a, b, c and d respectively. Prove that these lines meet in the point z given by $z = \dfrac{a^{-1} + b^{-1} - c^{-1} - d^{-1}}{a^{-1}b^{-1} - c^{-1}d^{-1}}$.

Solution:

Since P, Q, R are collinear, $\begin{vmatrix} z & \bar{z} & 1 \\ c & \bar{c} & 1 \\ d & \bar{d} & 1 \end{vmatrix} = 0$



$$\Rightarrow z(\bar{c} - \bar{d}) - \bar{z}(c - d) + (c\bar{d} - \bar{c}d) = 0 \quad \text{...(1)}$$

Similarly, $z(\bar{a} - \bar{b}) - \bar{z}(a - b) + (a\bar{b} - \bar{a}b) = 0 \quad \text{...(2)}$

From {(1) × (a – b)} – {(2) × (c – d)}, we get

$$z\left[(\bar{c} - \bar{d})(a - b) - (\bar{a} - \bar{b})(c - d)\right] = (a\bar{b} - b\bar{a})(c - d) - (c\bar{d} - \bar{c}d)(a - b)$$

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Now $\bar{a} = r^2 \Rightarrow \bar{a} = \dfrac{r^2}{a}$.

Similarly $\bar{b} = \dfrac{r^2}{b}$, $\bar{c} = \dfrac{r^2}{c}$, $\bar{d} = \dfrac{r^2}{d}$.

From (3), $z\left[\left(\dfrac{r^2}{c} - \dfrac{r^2}{d}\right)(a - b) - \left(\dfrac{r^2}{a} - \dfrac{r^2}{b}\right)(c - d)\right] = \left(\dfrac{ar^2}{b} - \dfrac{br^2}{a}\right)(c - d) - \left(\dfrac{cr^2}{d} - \dfrac{r^2}{c}d\right)(a - b)$

$$\Rightarrow z\left[\frac{(d - c)(a - b)}{cd} - \frac{(b - a)(c - d)}{ab}\right] = \frac{(a^2 - b^2)(c - d)}{ab} - \frac{(c^2 - d^2)(a - b)}{cd}$$

$$\Rightarrow z\left[-\frac{1}{cd} + \frac{1}{ab}\right] = \frac{a + b}{ab} - \frac{c + d}{cd} \Rightarrow z = \frac{a^{-1} + b^{-1} - c^{-1} - d^{-1}}{a^{-1}b^{-1} - c^{-1}d^{-1}}$$

Illustration 56

Find the equation of the circle which touches the line $iz + \bar{z} + 1 + i = 0$ and has the lines $(1 - i)z = (1 + i)\bar{z}$ and $(1 + i)z + (i - 1)\bar{z} = 4i$ as its normals.

Solution:

Clearly point of intersection of normal would be the centre of the required circle.

$$(1 - i)z = (1 + i)\bar{z} \Rightarrow \frac{z}{\bar{z}} = \frac{1 + i}{1 - i} \times \frac{1 + i}{1 + i} = \frac{zi}{z} = i \Rightarrow z = i\bar{z}$$

Putting in $(1 + i)z + (i - 1)\bar{z} = 4i$, we get $z = 1 + i$

Now equation of tangent can be rewritten as,

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$(1-i)(iz+\bar{z}+1+i)=0 \Rightarrow \quad (1+i)z+(1-i)\bar{z}+2=0$

Now distance of $z=(1+i)$ tangent is radius

$$\Rightarrow r=\frac{|(1+i)(1+i)+(1-i)(1-i)+2|}{2|1-i|}=\frac{1}{\sqrt{2}}$$

Thus the equation of required circle is $|z-(1+i)|=\frac{1}{\sqrt{2}}$

11. GEOMETRICAL APPLICATIONS OF ARGUMENT:

Locus of z satisfying $\arg(z-z_1)=\alpha$ is ray with vertex at z_1 and making angle 'α' with +ve direction of real axis (Locus excludes the point z_1)

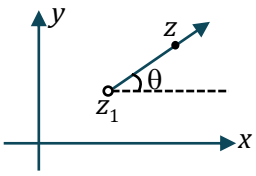


Illustration 57

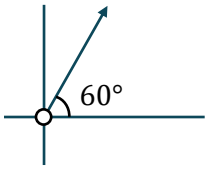
Let amp(z) denote the principle value of argument. Sketch the locus of z satisfying the following conditions :

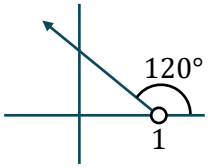
(i) $\text{amp}(z)=\frac{\pi}{3}$ (ii) $\text{amp}(z-1)=\frac{2\pi}{3}$ (iii) $\text{amp}(z-2-3i)=0$

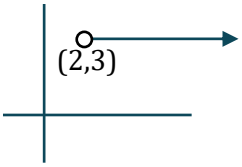
(iv) $\text{amp}(z+2+i)=-\frac{\pi}{2}$ (v) $\text{amp}(z+6i)=\frac{-2\pi}{3}$

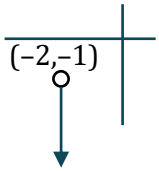
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Solution:

(i) 

(ii) 

(iii) 

(iv) 

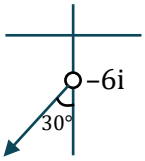
(v) 

Illustration 58

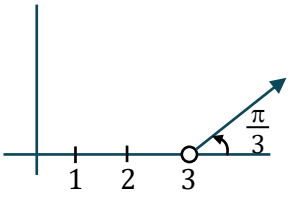
Let arg(z) denote the principle value of argument. Plot the locus of z satisfying following conditions :

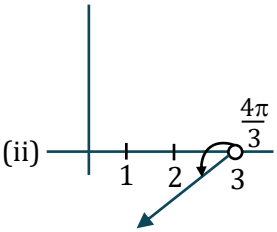
(i) $\arg(z-3)=\frac{\pi}{3}$ (ii) $\arg(z-3)=\frac{4\pi}{3}$

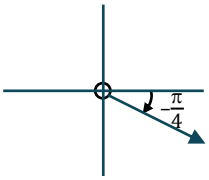
(iii) $\arg(z)=-\frac{\pi}{4}$ (iv) $\arg(z-2i)=\frac{3\pi}{4}$

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Solution:

(i) 

(ii) 

(iii) 

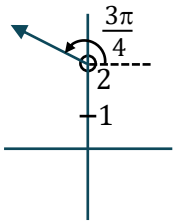
(iv) 

Illustration 59

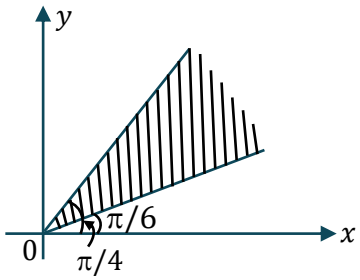
Let amp(z) denote the principle value of argument. Plot the locus of z satisfying following conditions :

(i) $\frac{\pi}{6} < \text{amp}(z) < \frac{\pi}{4}$ (ii) $\frac{\pi}{6} < \text{amp}(z+1) < \frac{\pi}{3}$

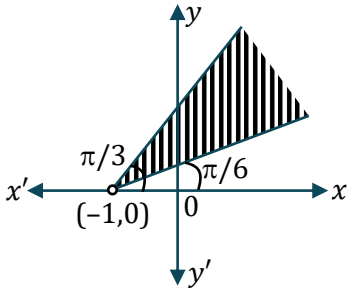
Solution:

(i) We observe that $\frac{\pi}{6} < \text{amp}(z) < \frac{\pi}{4}$ represents the set of all points, except the origin, lying between the lines passing through the origin and making 30° and 45° with the positive direction of x-axis

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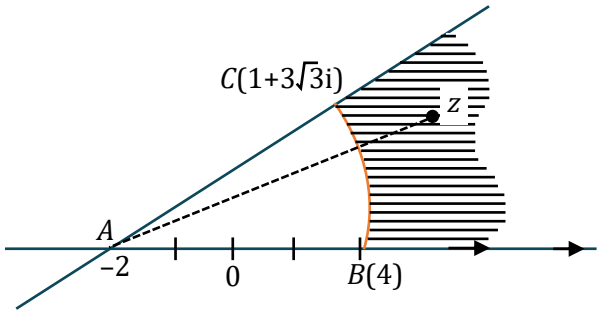
(ii) The region represented by $\frac{\pi}{6} < \text{amp}(z+1) < \frac{\pi}{3}$ is shown below. The point (–1, 0) is not included in the shaded region.



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Illustration 60

Let amp(z) denote the principle value of argument. Shaded region in the given figure is represented by which of the following options -



(A) $|z+2| \geq 6, 0 \leq \text{amp}(z) \leq \frac{\pi}{6}$

(B) $|z+2| \leq 6, 0 \leq \text{amp}(z+2) \leq \frac{\pi}{3}$

(C) $|z+2| \geq 6, 0 \leq \text{amp}\,(z+2) \leq \frac{\pi}{3}$

(D) $|z+2| \leq 6, 0 \leq \text{amp}(z+2) \leq \frac{\pi}{3}$

Solution:

Note that $AB = 6$ and $1 + 3\sqrt{3}i = -2 + 3 + 3\sqrt{3}i$

$= -2 + 6\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) = -2 + 6\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$

$\angle BAC = \frac{\pi}{3}$

Thus, shaded region is given by $|z+2| \geq 6$ and $0 \leq \arg(z+2) \leq \frac{\pi}{3}$ **Ans. (C)**

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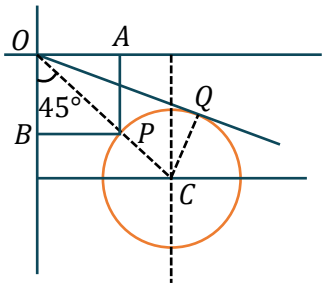
Illustration 61

Find z such that $|z - 2 + 2i| \leq 1$ and z has

(i) least modulus

(ii) numerically least amplitude

Solution:



(i) Given equation represents a circle with centre (2, –2) and radius 1. Distance of $|z - 2 + 2i| = 1$ will be minimum from origin along the line which is normal to the circle passing through the origin, so it will pass through the centre of the circle also.

$OP = \left(2\sqrt{2} - 1\right)$ $OA = OP\cos45°$ and $OB = OP\sin45°$ so that

$$z = \frac{\left(2\sqrt{2}-1\right)}{\sqrt{2}} - i\left(\frac{2\sqrt{2}-1}{\sqrt{2}}\right)$$

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(ii) We have to find the complex numbers z represent by Q where OQ is a tangent to the circle Now

$$OQ = \sqrt{OC^2 - CQ^2} = |2-2i|^2 = 2\sqrt{2^2 + (-2)^2 - 1} = \sqrt{7}$$

$\therefore |z| = \sqrt{7}$

$$\angle QOX = -|\angle COX - \angle COQ| = -\left|\frac{\pi}{4} - \sin^{-1}\frac{1}{OC}\right|$$

$$= -\left|\frac{\pi}{4} - \sin^{-1}\frac{1}{|2-2i|}\right| = -\left(\frac{\pi}{4} - \sin^{-1}\frac{1}{2\sqrt{2}}\right)$$

$$\Rightarrow \text{amp}z = -\left(\frac{\pi}{4} - \sin^{-1}\frac{1}{2\sqrt{2}}\right)$$

$\Rightarrow z = |z|\{\cos(\text{amp}z) + i\sin(\text{amp}z)\}$

$$= \sqrt{7}\left[\cos\left\{-\left(\frac{\pi}{4} - \sin^{-1}\frac{1}{2\sqrt{2}}\right)\right\} + i\sin\left\{-\left(\frac{\pi}{4} - \sin^{-1}\frac{1}{2\sqrt{2}}\right)\right\}\right]$$

$$= \sqrt{7}\left[\cos\left(\frac{\pi}{4} - \sin^{-1}\frac{1}{2\sqrt{2}}\right) - i\sin\left(\frac{\pi}{4} - \sin^{-1}\frac{1}{2-\sqrt{2}}\right)\right].$$

Illustration 62

Among the complex numbers z which satisfies $|z - 25i| \le 15$, find the complex numbers z having

(a) least positive argument (b) maximum positive argument

(c) least modulus (d) maximum modulus

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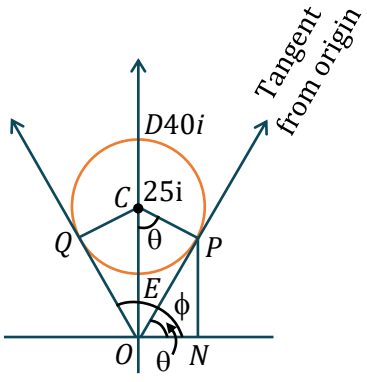
Solution:

The complex numbers z satisfying the condition $|z - 25i| \le 15$ are represented by the points inside and on the circle of radius 15 and centre at the point $C(0, 25)$.

The complex number having least positive argument and maximum positive arguments in this region are the points of contact of tangents drawn from origin to the circle

Here θ = least positive argument

and ϕ = maximum positive argument



$\therefore$ In $\Delta OCP, OP = \sqrt{(OC)^2 - (CP)^2} = \sqrt{(25)^2 - (15)^2} = 20$

and $\sin\theta = \frac{OP}{OC} = \frac{20}{25} = \frac{4}{5}$

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$\therefore \tan\theta = \frac{4}{3} \Rightarrow \theta = \tan^{-1}\left(\frac{4}{3}\right)$

Thus, complex number at P has modulus 20 and argument $\theta = \tan^{-1}\left(\frac{4}{3}\right)$

$\therefore z_p = 20(\cos\theta + i\sin\theta) = 20\left(\frac{3}{5} + i\frac{4}{5}\right)$

$\therefore z_p = 12 + 16i$

Similarly z_Q = –12 + 16i

From the figure, E is the point with least modulus and D is the point with maximum modulus.

Hence, $z_E = \overrightarrow{OE} = \overrightarrow{OC} - \overrightarrow{EC} = 25i - 15i = 10i$

and $z_D = \overrightarrow{OD} = \overrightarrow{OC} + \overrightarrow{CD} = 25i + 15i = 40i$

Illustration 63

If z is a point on the Argand plane such that $|z - 1| = 1$, then prove that $\frac{z-2}{z} = i\tan(\arg z)$.

Solution:

Since $|z - 1| = 1$,

$\therefore$ Let $z - 1 = \cos\theta + i\sin\theta$

Then, $z - 2 = \cos\theta + i\sin\theta - 1$

$= -2\sin^2\frac{\theta}{2} + 2i\sin\frac{\theta}{2}\cos\frac{\theta}{2} = 2i\sin\frac{\theta}{2}\left(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2}\right)$ ...(i)

and $z = 1 + \cos\theta + i\sin\theta$

$= 2\cos^2\frac{\theta}{2} + 2i\sin\frac{\theta}{2}\cos\frac{\theta}{2} = 2\cos\frac{\theta}{2}\left(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2}\right)$ ...(ii)

From (i) and (ii), we get $\frac{z-2}{z} = i\tan\frac{\theta}{2} = i\tan(\arg z)$ $\left(\because \arg z = \frac{\theta}{2} \text{ from (ii)}\right)$

Illustration 64

Let amp(z) denote the principle value of argument. If $|z| \le 1$, $|w| \le 1$, show that

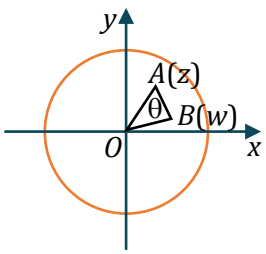
$|z - w|^2 \le (|z| - |w|)^2 + (\text{amp}(z) - \text{amp}(w))^2$

Solution:

Let us consider a unit circle with its centre as the origin

Let $\angle AOX = \theta_1$ and $\angle BOX = \theta_2$

$\therefore$ amp(z) = θ_1 and amp(w) = θ_2



Now in ΔOAB,

$\Rightarrow \cos\theta = \frac{|\overline{OA}|^2 + |\overline{OB}|^2 - |\overline{BA}|^2}{2|\overline{OA}||\overline{OB}|^2}$

$\Rightarrow |\overline{BA}|^2 = |\overline{OA}|^2 + |\overline{OB}|^2 - 2|\overline{OA}||\overline{OB}|\cos\theta$

$\Rightarrow |z - w|^2 = |z|^2 + |w|^2 - 2|z||w|\cos\theta = (|z| - |w|)^2 - 4|z||w|\sin^2\left(\frac{\theta}{2}\right)$

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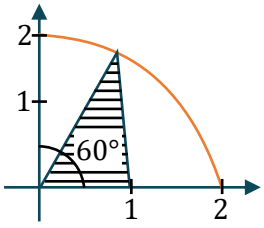
We know that $\sin\frac{\theta}{2} \le \frac{\theta}{2}$

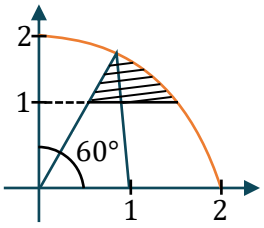
$\Rightarrow \left(\frac{\theta}{2}\right)^2 \ge \sin^2\frac{\theta}{2} \Rightarrow \left(\frac{\theta}{2}\right)^2 \ge |z||w|\sin^2\frac{\theta}{2}$ [since $|z|, |w| \le 1$, it is a unit circle]

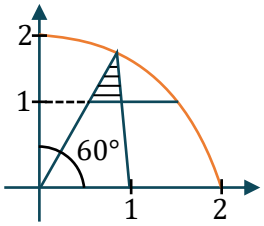
Hence $|z - w|^2 \le (|z| - |w|)^2 + (\text{amp}(z) - \text{amp}(w))^2$.

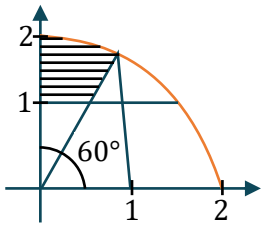
Illustration 65

Let amp(z) denote the principle value of argument. The region represented by inequalities amp $z \le \frac{\pi}{3}$; $|z| \le 2$; Im(z) ≥ 1 in the Argand diagram is given by -

(A) 

(B) 

(C) 

(D) 

Ans. (B)

Solution:

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Illustration 66

Let amp(z) denote the principle value of argument. Prove the following inequalities geometrically and analytically :

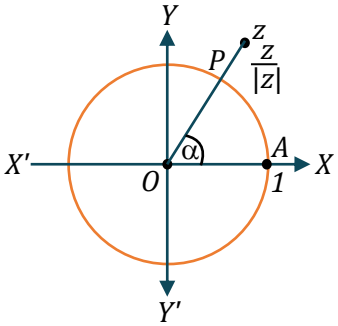
(a) $\left|\frac{z}{|z|} - 1\right| \le |\text{amp}z|$ **(b)** $|z - 1| \le ||z| - 1| + |z|\,|\text{amp}\,z|$

Solution:

(a) The quantity $\frac{z}{|z|}$ lies on a unit circle centred at the origin

It is clear that $\left|\frac{z}{|z|} - 1\right| = AP \le \text{arc}(AP) = \alpha = \text{amp}\,z$

If α is negative, this gives $\left|\frac{z}{|z|} - 1\right| \le |\arg z|$



To show this analytically, put

$z = r(\cos\alpha + i\sin\alpha)$

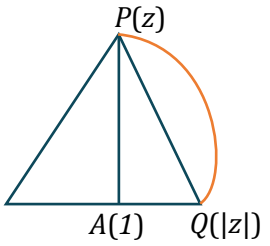
where $r = |z|$

Then $\left|\frac{z}{|z|} - 1\right| = |(\cos\alpha - 1) + i\sin\alpha|$

$= [(\cos\alpha - 1)^2 + \sin^2\alpha]^{1/2} = [2(1 - \cos\alpha)]^{1/2}$

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(b) Referring to the shown figure and using the result of part (a), we have



$|z-1| = AP \le AQ + QP = ||z|-1| + |z-|z||$

$\Rightarrow |z-1| \le ||z|-1| + |z|\left|\frac{z}{|z|}-1\right| \le ||z|-1| + |z||\arg z|$

Analytically, we can get this result as follows :

$|z-1| = |(z-|z|) + (|z|-1)| \le |z-|z|| + ||z|-1|$

$\Rightarrow |z-1| \le ||z|-1| + |z|\left|\frac{z}{|z|}-1\right| \le ||z|-1| + |z||\arg z|$

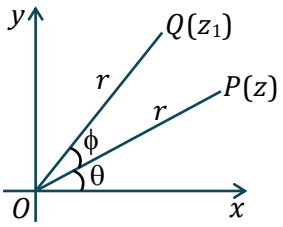
12. ROTATION IN COMPLEX NUMBERS:

12.1 Rotation w.r.t. Origin:

Let point P represents a complex number z in Argand Plane with $|z| = r, \arg(z) = \theta$. Then $z \cdot e^{i\phi} = r \cdot e^{i\theta} \cdot e^{i\phi} = r \cdot e^{i(\theta+\phi)}$

Observe that multiplying z by $e^{i\phi}$ has no effect on modulus of z, but increases its argument by angle ϕ.

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Hence, multiplying z by $e^{i\phi}$ rotates its position vector by angle ϕ with respect to origin(see fig.).

Note:

(i) Multiplying z by i rotates its position vector by 90° in anticlockwise direction w.r.t origin.

(ii) If $\overrightarrow{OP}$ and $\overrightarrow{OQ}$ are of unequal magnitude then we can use rotation of unit vectors i.e. if we multiply unit vector OP by $e^{i\phi}$ rotates it by an angle ϕ with respect to origin and thus $OPe^{i\phi}$ is a unit vector OQ

$\Rightarrow OQ = OPe^{i\phi} \Rightarrow \frac{z_1}{|z_1|} = \frac{z}{|z|}e^{i\phi}$.

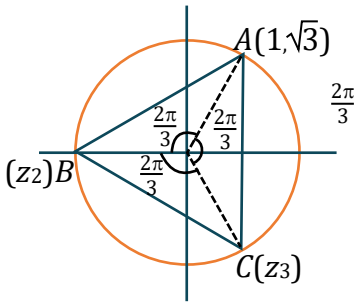
Illustration 67

Suppose z_1, z_2, z_3 are the vertices of an equilateral triangle (taken anticlockwise) inscribed in the circle $|z| = 2$. If $z_1 = 1 + i\sqrt{3}$, then find z_2 and z_3.

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Solution:

Clearly $\angle AOB = \frac{2\pi}{3}$. So, if we rotate vector $\overrightarrow{OA}$ by angle $\frac{2\pi}{3}$ in anticlockwise direction, then we must get vector $\overrightarrow{OB}$.



Thus, we get

$$\overrightarrow{OA}\cdot e^{i\frac{2\pi}{3}} = \overrightarrow{OB} \Rightarrow z_2 = z_1 \cdot e^{i\frac{2\pi}{3}} \Rightarrow z_2 = \left(1+i\sqrt{3}\right)\cdot\left(\frac{-1+i\sqrt{3}}{2}\right) = -2+0i$$

Similarly by rotating $\overrightarrow{OA}$ by angle $\frac{2\pi}{3}$ in clockwise direction, then we will get vector $\overrightarrow{OC}$ i.e.

$$\overrightarrow{OC} = \overrightarrow{OA}\cdot e^{-i\frac{2\pi}{3}} \Rightarrow z_3 = z_1 \cdot e^{-i\frac{2\pi}{3}} \Rightarrow z_2 = \left(1+i\sqrt{3}\right)\cdot\left(\frac{-1-i\sqrt{3}}{2}\right) = 1-i\sqrt{3}$$

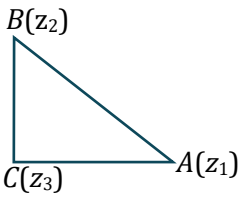
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Illustration 68

Triangle ABC is right angled at C, with $CB = AC$. If vertices of triangle are taken in anticlockwise sense, $A(1, 2)$ and $C(-1, 3)$, then find the coordinates of point B.

Solution:

Let complex number z denote the point B. $CA = CB$ means by rotating vector $\overrightarrow{CA}$ by 90°, we must get vector $\overrightarrow{CB}$.



Thus, applying rotation at point C, we have

$$\overrightarrow{CA}\cdot e^{i\frac{\pi}{2}} = \overrightarrow{CB} \Rightarrow z+1-3i = (2-i)\cdot i \Rightarrow z = 5i$$

Hence, point B is $(0, 5)$.

Illustration 69

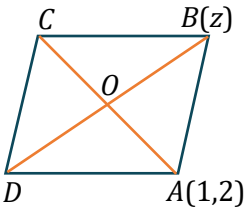
Vertices of rhombus $ABCD$ whose diagonals intersect at origin are taken in anticlockwise sense with $A(1, 2)$. If area of rhombus is 20 square units, then find coordinates of points B, C and D.

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Solution:

Let complex number z denote the point B.

Area of Rhombus $= \frac{1}{2}(AC)(BD) = 2(OA)(OB) = 2\sqrt{5}\cdot OB \Rightarrow OB = 2\sqrt{5}$



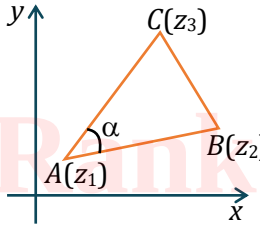
So by rotating unit vector OA by angle $\frac{\pi}{2}$ anti-clockwise, we will get unit vector OB i.e.

$OB = OA\cdot e^{i\frac{\pi}{2}} \Rightarrow z = (1+2i)\cdot i = i-2$

Hence, coordinates of point B are (–2, 1). Now coordinates of C and D can be found easily by using origin as the mid-point of AC and BD i.e. $C(-1,-2)$ and $D(2,-1)$.

12.2 Rotation w.r.t Any Point:

In general, if z_1, z_2, z_3 be the vertices of ΔABC taken in anti-clockwise sense,



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then $\frac{z_3-z_1}{z_2-z_1} = \frac{|z_3-z_1|}{|z_2-z_1|}e^{i\alpha}$.

Here $\arg\left(\frac{z_3-z_1}{z_2-z_1}\right) = \alpha$.

Illustration 70

Triangle ABC is right angled at C, with $CB = AC$. If vertices of triangle are taken in anticlockwise sense, $A(1, 2)$ and $B(-1, 3)$, then find the coordinates of point C without using rotation at point C.

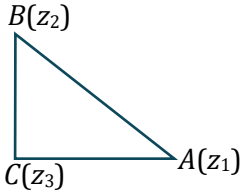
Solution:

Let complex number z denote the point C. Clearly $B = \frac{\pi}{4}$. So by rotating unit vector BA by angle $\frac{\pi}{4}$ clockwise, we will get unit vector BC.

Thus, applying rotation at point B, we have

$BC = BA\cdot e^{-i\frac{\pi}{4}} \Rightarrow \frac{\overrightarrow{BC}}{BC} = \frac{\overrightarrow{BA}}{BA}\cdot e^{-i\frac{\pi}{4}} \Rightarrow \overrightarrow{BC} = \frac{BC}{BA}\cdot\overrightarrow{BA}\cdot e^{-i\frac{\pi}{4}}$

$\Rightarrow \overrightarrow{BC} = \frac{1}{\sqrt{2}}\cdot\overrightarrow{BA}\cdot e^{-i\frac{\pi}{4}} \Rightarrow z+1-3i = \frac{1}{\sqrt{2}}(2-i)\frac{1-i}{\sqrt{2}} \Rightarrow z = \frac{-1+3i}{2}$



Hence, point B is $C\left(\frac{-1}{2},\frac{3}{2}\right)$.

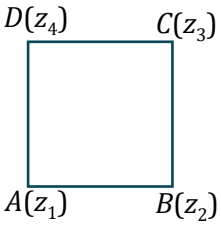
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Illustration 71

If the vertices of a square $ABCD$ are z_1, z_2, z_3 & z_4 in anticlockwise sense then find z_3 & z_4 in terms of z_1 & z_2.

Solution:

Method-I : Using vector rotation at A



$\overrightarrow{AD} = \overrightarrow{AB}.e^{i\frac{\pi}{2}}$

$\Rightarrow z_4 - z_1 = (z_2 - z_1)i$

$\Rightarrow z_4 = (1 - i)\, z_1 + iz_2$

Similarly, using vector rotation at B, we get

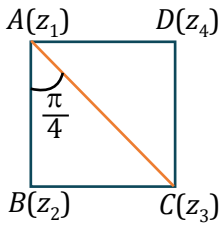
$\overrightarrow{BC} = \overrightarrow{BA}.e^{-i\frac{\pi}{2}}$

$\Rightarrow z_3 - z_2 = (z_1 - z_2)(-i)$

$\Rightarrow z_3 = (1 + i)z_2 - iz_1$

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Method-II : Using unit vector rotation at angle A



$$\frac{z_3 - z_1}{|z_3 - z_1|} = \frac{z_2 - z_1}{|z_2 - z_1|}e^{i\frac{\pi}{4}} \Rightarrow \frac{z_3 - z_1}{z_2 - z_1} = \frac{|z_3 - z_1|}{|z_2 - z_1|}e^{i\frac{\pi}{4}} = \frac{AC}{AB}.e^{i\frac{\pi}{4}} = \sqrt{2}e^{i\frac{\pi}{4}}$$

$$[\because] AC = \sqrt{2}\,AB \Rightarrow \frac{z_3 - z_1}{z_2 - z_1} = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right) \Rightarrow z_3 - z_1 = (z_2 - z_1)(1 + i)$$

$\Rightarrow z_3 = z_1 + (z_2 - z_1)(1 + i)$

Similarly we can prove that $z_4 = z_2 + (1 - i)(z_1 - z_2)$

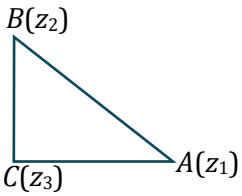
Illustration 72

Complex numbers z_1, z_2, z_3 are the vertices A, B, C respectively of an isosceles right angled triangle with right angle at C. Show that $(z_1 - z_2)^2 = 2(z_1 - z_3)(z_3 - z_2)$.

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Solution:

In the isosceles triangle ABC, $AC = BC$ and $BC \perp AC$. It means that AC is rotated through angle $\pi/2$ to occupy the position BC.



Hence we have, $\frac{z_2 - z_3}{z_1 - z_3} = e^{\pm\frac{i\pi}{2}} = \pm i \Rightarrow z_2 - z_3 = \pm i(z_1 - z_3)$

$\Rightarrow z_2^2 + z_3^2 - 2z_2z_3 = -\left(z_1^2 + z_3^2 - 2z_1z_3\right)$

$\Rightarrow z_1^2 + z_2^2 - 2z_1z_2 = 2z_1z_3 + 2z_2z_3 - 2z_1z_2 - 2z_3^2 = 2\left(z_1 - z_3\right)\left(z_3 - z_2\right)$

$\Rightarrow \left(z_1 - z_2\right)^2 = 2\left(z_1 - z_3\right)\left(z_3 - z_2\right)$

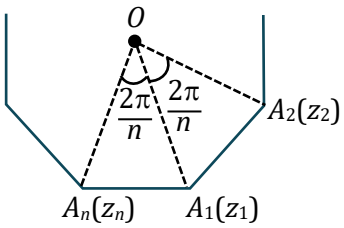
Illustration 73

Consider an 'n' sided regular polygon with the origin as its centre. If z_1 be the complex number representing a vertex A_1 of the polygon, find the complex numbers associated with the vertices that is adjacent to A_1.

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Solution:

Vertex adjacent to A_1 is either A_2 or A_n. Let z_2 and z_n represent the vertices A_2 and A_n respectively. Considering the rotation about the origin, we get



$\frac{z_2 - 0}{z_1 - 0} = \frac{|z_2|}{z_1} e^{i\frac{2\pi}{n}} = e^{i\frac{2\pi}{n}}$

$\Rightarrow z_2 = z_1 e^{i\frac{2\pi}{n}}$

Similarly, $\frac{z_n}{z_1} = e^{-i\frac{2\pi}{n}}$

$\Rightarrow z_n = z_1 e^{-i\frac{2\pi}{n}}$

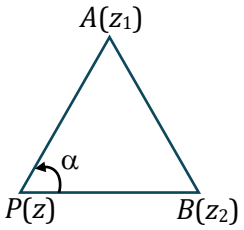
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12.3 Locus of Satisfying $\arg\left(\frac{z-z_1}{z-z_2}\right)=\alpha$, $\alpha > 0$

Let $A(z_1), B(z_2), P(z)$. Then $\arg\left(\frac{z-z_1}{z-z_2}\right)=\alpha \Rightarrow \frac{z-z_1}{z-z_2}=\left|\frac{z-z_1}{z-z_2}\right|\cdot e^{i\alpha} \Rightarrow \frac{z-z_1}{|z-z_1|}=\frac{z-z_2}{|z-z_2|}\cdot e^{i\alpha}$

Thus, point P must be located in such a way that unit vector AP *i.e.* $\frac{z-z_1}{|z-z_1|}$ must be obtained upon rotating unit vector BP *i.e.* $\frac{z-z_2}{|z-z_2|}$ by angle α in anti-clockwise sense.

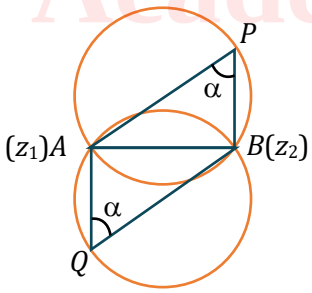
Hence angle between vectors $\overrightarrow{PA}$ *and* $\overrightarrow{PB}$ must be α i.e. $\angle APB=\alpha$. In other words point P moves such that segment AB always subtends angle α at P.



Thus P must lie on major (minor) arc of a circle in which AB is a chord, which subtends angle α at points on its circumference.

Note :

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(i) If z_1, z_2, z_3, z_4 are distinct complex numbers lying on a circle (in order), then $\frac{(z_1-z_2)(z_4-z_3)}{(z_1-z_4)(z_2-z_3)}$ is purely real.

(ii) If $P(z)$ lies on the are lying above AB as shown in figure then $\arg\left(\frac{z-z_1}{z-z_2}\right)=-\alpha$

(iii) If P lies on the are below AB as shown in figure, then $\arg\left(\frac{z-z_1}{z-z_2}\right)=\alpha$

Illustration 74

Plot the region represented by complex number z satisfying $\text{amp}\left(\frac{z+1}{z-1}\right)=\frac{\pi}{2}$ in the Argand plane. Here amp(z) denotes the principle value of argument.

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Solution:

The point z satisfies $\text{amp}\left(\frac{z+1}{z-1}\right)=\frac{\pi}{2}$ means that segment joining $A(1,0)$ and $B(-1,0)$ subtends a right angle at $P(z)$. Clearly P must lie on circumference of a circle with A and B as ends of diameter. If $P(z)$ lies above AB, then $\text{amp}\left(\frac{z+1}{z-1}\right)=\frac{-\pi}{2}$ and if $P(z)$ lies below AB, then $\text{amp}\left(\frac{z+1}{z-1}\right)=\frac{\pi}{2}$. Hence locus of z is the semicircle shown in figure which lies below the chord AB (excluding A and B).

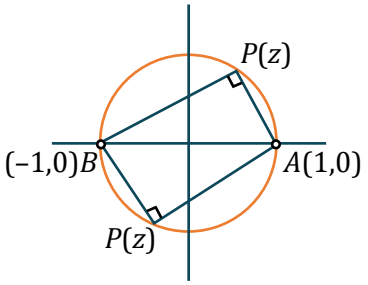
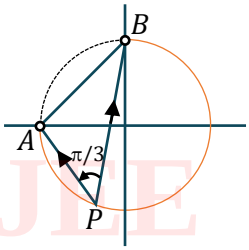


Illustration 75

Plot the region represented by complex number z satisfying $\text{amp}\left(\frac{z+1}{z-i}\right)=\frac{\pi}{3}$ in the Argand plane. Here amp(z) denotes the principle value of argument.

Solution:

The point z satisfies $\text{amp}\left(\frac{z+1}{z-i}\right)=\frac{\pi}{3}$ means that segment joining subtends 60° at $P(z)$. Clearly P must lie on circumference of a circle with AB as chord subtending 60° on its circumference and this angle must be obtained by rotating vector $\overrightarrow{PB}$ towards $\overrightarrow{PA}$.



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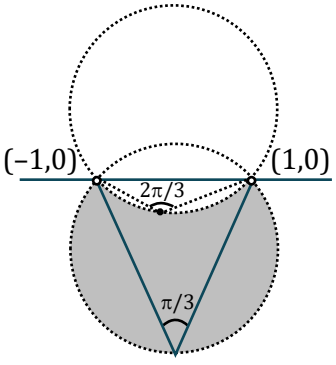
Hence locus of z is the major arc of circle shown in figure which lies below the chord AB (excluding A and B), where AB subtends 60° at every point on it.

Illustration 76

Plot the region represented by complex number z satisfying $\frac{\pi}{3}\le\text{amp}\left(\frac{z+1}{z-1}\right)\le\frac{2\pi}{3}$ in the Argand plane. Here amp(z) denotes the principle value of argument.

Solution:

Let us take $\text{amp}\left(\frac{z_1+1}{z_1-1}\right)=\frac{2\pi}{3}$. Clearly z lies on the minor arc of the circle passing through (1, 0) and (–1, 0). Similarly, $\text{amp}\left(\frac{z+1}{z-1}\right)=\frac{\pi}{3}$ means that 'z' is lying on the major arc of the circle passing through (1, 0) and (–1, 0). Now if we take any point in the region included between two arcs say $P_1(z_1)$ we get $\frac{\pi}{3}\le\text{amp}\left(\frac{z+1}{z-1}\right)\le\frac{2\pi}{3}$.



Thus $\frac{\pi}{3}\le\text{amp}\left(\frac{z+1}{z-1}\right)\le\frac{2\pi}{3}$ represents the shaded region (excluding points (1, 0) and (–1, 0)).

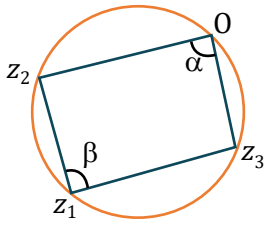
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Illustration 77

If z_1, z_2, z_3 are complex numbers such that $\frac{2}{z_1}=\frac{1}{z_2}+\frac{1}{z_3}$, show that the points represented by z_1, z_2, z_3 lie on a circle passing through the origin.

Solution:

We have, $\frac{2}{z_1}=\frac{1}{z_2}+\frac{1}{z_3} \Rightarrow \frac{1}{z_1}-\frac{1}{z_2}=\frac{1}{z_3}-\frac{1}{z_1} \Rightarrow \frac{z_2-z_1}{z_1z_2}=\frac{z_1-z_3}{z_1z_3}$



$\Rightarrow \frac{z_2-z_1}{z_3-z_1}=\frac{-z_2}{z_3} \Rightarrow \arg\left(\frac{z_2-z_1}{z_3-z_1}\right)=\arg\left(\frac{-z_2}{z_3}\right)$

$\arg\left(\frac{z_2-z_1}{z_3-z_1}\right)=\pi+\arg\left(\frac{z_2}{z_3}\right) \Rightarrow \text{or } \beta=\pi-\arg\frac{z_3}{z_2}=\pi-\alpha \Rightarrow \alpha+\beta=\pi$

Thus the sum of a pair of opposite angles of the quadrilateral is 180°. Hence, the points 0, z_1, z_2 and z_3 are the vertices of a cyclic quadrilateral i.e. lie on a circle.

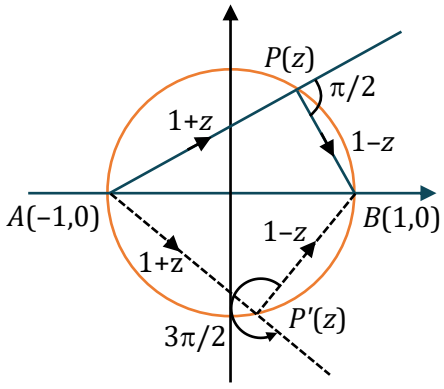
Illustration 78

If $|z| = 1$, then prove that the points represented by $\sqrt{\frac{1+z}{1-z}}$ lie on one or other of two fixed perpendicular straight lines.

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Solution:

z lies on a unit circle having centre at the origin.



$amp\left(\frac{1+z}{1-z}\right)=\pm\frac{\pi}{2} \Rightarrow \frac{1+z}{1-z}=ke^{\pm i\frac{\pi}{2}}$;

where k is a real parameter and its value depends upon the position of z.

Let $\alpha=\sqrt{\frac{1+z}{1-z}}$

$\Rightarrow \alpha=\sqrt{k}\,e^{i\frac{\pi}{4}}$ or $\sqrt{k}\,e^{i\frac{3\pi}{4}}$

$\Rightarrow \alpha$ lies on one or other of the two perpendicular lines.

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13. k^{th} ROOT OF *A* COMPLEX NUMBER:

For any real θ and positive integer k, we have

$$(\cos\theta+i\sin\theta)^{\frac{1}{k}}=\cos\frac{2n\pi+\theta}{k}+i\sin\frac{2n\pi+\theta}{k};n=0,1,2,...,(k-1)$$

Note: In case n is a rational number i.e. $n=\frac{p}{q};p,q\in\mathbb{Z},q\neq0,\gcd(p,q)=1$, then $(\cos\theta+i\sin\theta)^{\frac{p}{q}}$ has 'q' different values given by

$$(\cos\theta+i\sin\theta)^{\frac{p}{q}}=\cos\frac{(2n\pi+\theta)p}{q}+i\sin\frac{(2n\pi+\theta)p}{q};n=0,1,2,...,(q-1).$$

Illustration 79

Solve $z^5=16\sqrt{2}(1+i)$ and express the solutions in exponential form.

Solution:

$$z^5=32\cdot e^{\frac{i\pi}{4}}\Rightarrow z^5=32\cdot e^{i\left(2n\pi+\frac{\pi}{4}\right)}\Rightarrow z^5=32\cdot e^{i(8n+1)\frac{\pi}{4}};\ n\in\mathbb{Z}$$

$$\Rightarrow\ z=2e^{i(8n+1)\frac{\pi}{20}}=2e^{i\frac{\pi}{20}},\ 2e^{i\frac{9\pi}{20}},\ 2e^{i\frac{17\pi}{20}},\ 2e^{i\frac{25\pi}{20}},\ 2e^{i\frac{33\pi}{20}}\ \text{(Put } k=0,1,2,3,4)$$

13.1 Cube Roots Of Unity :

Cube roots of unity $(\sqrt[3]{1})$ are roots of equation $z^3=1$.

Factorising, we get $(z-1)(z^2+z+1)=0$

$\Rightarrow z=1$ or $z^2+z+1=0$

$$\Rightarrow z=1,\ \frac{-1+i\sqrt{3}}{2},\ \frac{-1-i\sqrt{3}}{2}$$

$$\Rightarrow z=1,\ e^{i\frac{2\pi}{3}},\ e^{i\frac{4\pi}{3}}$$

Following points must be noted :

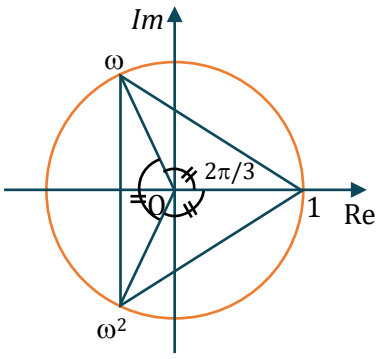
(a) Non-real cube roots of unity are denoted by ω, ω^2.

(b) $1+\omega+\omega^2=0$ and $\omega^3=1$

ω, ω^2 are the roots of $z^2+z+1=0\Rightarrow\omega+\omega^2=-1$ and $\omega.\omega^2=1$

(c) If $n\in\mathbb{Z}$, then $\omega^3=1\Rightarrow(\omega^3)^n=1$

$\Rightarrow \omega^{3n}=1, \omega^{3n+1}=\omega, \omega^{3n+2}=\omega^2$



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(d) $1 + \omega^n + \omega^{2n} = \begin{cases} 3 & ; \quad n = 3k;\ k \in \mathbb{Z} \\ 0 & ; \quad n \neq 3k;\ k \in \mathbb{Z} \end{cases}$

(e) Geometrically 1, ω, ω^2 are vertices of an equilateral triangle centered at (0, 0)

(f) The following factorisations should be remembered:

$x^3 - 1 = (x - 1)(x - \omega)(x - \omega^2)$;

$a^3 - b^3 = (a - b)(a - \omega b)(a - \omega^2 b)$

$a^3 + b^3 = (a + b)(a + \omega b)(a + \omega^2 b)$;

$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a + \omega b + \omega^2 c)(a + \omega^2 b + \omega c)$

Illustration 80

If $n \in \mathbb{Z}$ and α, β are non real cube roots of unity, then prove that $\alpha^n + \beta^n = 2\cos\left(\frac{2n\pi}{3}\right)$

Solution:

$\alpha = \cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}$; $\beta = \cos\frac{2\pi}{3} - i\sin\frac{2\pi}{3}$

$\therefore\ \alpha^n + \beta^n = \left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)^n + \left(\cos\frac{2\pi}{3} - i\sin\frac{2\pi}{3}\right)^n$

$= \left(\cos\frac{2n\pi}{3} + i\sin\frac{2n\pi}{3}\right) + \left(\cos\frac{2n\pi}{3} - i\sin\left(\frac{2n\pi}{3}\right)\right) = 2\cos\left(\frac{2n\pi}{3}\right)$

Illustration 81

If α, β, γ are roots of $x^3 - 3x^2 + 3x + 7 = 0$, then find the value of $\frac{\alpha-1}{\beta-1} + \frac{\beta-1}{\gamma-1} + \frac{\gamma-1}{\alpha-1}$ using cube roots of unity.

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Solution:

We have $x^3 - 3x^2 + 3x + 7 = 0 \Rightarrow (x - 1)^3 + 8 = 0 \Rightarrow (x - 1)^3 = (-2)^3$

$\Rightarrow \left(\frac{x-1}{-2}\right)^3 = 1 \quad \Rightarrow \quad \frac{x-1}{-2} = (1)^{1/3} = 1, \omega, \omega^2$ (cube roots of unity)

$\therefore \quad x = -1, 1 - 2\omega, 1 - 2\omega^2$

Here $\alpha = -1, \beta = 1 - 2\omega, \gamma = 1 - 2\omega^2$

$\therefore \quad \alpha - 1 = -2, \beta - 1 = -2\omega, \gamma - 1 = -2\omega^2$

Then $\frac{\alpha-1}{\beta-1} + \frac{\beta-1}{\gamma-1} + \frac{\gamma-1}{\alpha-1} = \left(\frac{-2}{-2\omega}\right) + \left(\frac{-2\omega}{-2\omega^2}\right) + \left(\frac{-2\omega^2}{-2}\right) = \frac{1}{\omega} + \frac{1}{\omega} + \omega^2 = \omega^2 + \omega^2 + \omega^2$

Therefore $\frac{\alpha-1}{\beta-1} + \frac{\beta-1}{\gamma-1} + \frac{\gamma-1}{\alpha-1} = 3\omega^2$.

Illustration 82

Consider a triangle formed by the points $A\left(\frac{2}{\sqrt{3}}e^{i\frac{\pi}{2}}\right), B\left(\frac{2}{\sqrt{3}}e^{-i\frac{\pi}{6}}\right), C\left(\frac{2}{\sqrt{3}}e^{-i\frac{5\pi}{6}}\right)$. Then

(a) Find the complex number z', if roots of equation $z^3 = z'$ are complex numbers representing vertices of given triangle.

(b) Let $P(z)$ be any point on in-circle of given triangle ABC. Then prove that $AP^2 + BP^2 + CP^2 = 5$.

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Solution:

Let $z_1 = \frac{2}{\sqrt{3}}e^{i\frac{\pi}{2}}, z_2 = \frac{2}{\sqrt{3}}e^{-i\frac{\pi}{6}}, z_3 = \frac{2}{\sqrt{3}}e^{-i\frac{5\pi}{6}}$

Then $z_2 = z_3 \cdot e^{i\frac{2\pi}{3}}$ and $z_1 = z_2 \cdot e^{i\frac{2\pi}{3}}$ i.e. $z_2 = z_3\omega$ and $z_1 = z_3\omega^2$, where ω denotes a non-real cube root of unity.

Thus z_1, z_2, z_3 must be cube roots of z_3 i.e. roots of equation $z^3 = z_3$

Again, $z_2 = z_3 \cdot e^{i\frac{2\pi}{3}}$ means that z_2 is obtained by rotating z_3 by 120° w.r.t. origin, and $z_1 = z_2 \cdot e^{i\frac{2\pi}{3}}$ means that z_1 is obtained by rotating z_2 by 120°.

Thus AB, BC, CA subtend 120° at origin, which means triangle ABC is equilateral and inscribed in circle $|z| = \frac{2}{\sqrt{3}}$.

$\Rightarrow$ Inradius = $\frac{R}{2} = 1$

$\Rightarrow$ Equation of in circle is $|z| = \frac{1}{\sqrt{3}}$

Let $P(z)$ be any point on the in circle i.e. $z = \frac{1}{\sqrt{3}}e^{i\theta}$.

Then, $AP^2 = |z - z_1|^2 = |z|^2 + |z_1|^2 - (z\bar{z}_1 + \bar{z}z_1)$

$BP^2 = |z|^2 + |z_2|^2 - (z\bar{z}_2 + \bar{z}z_2)$ and $CP^2 = |z|^2 + |z_3|^2 - (z\bar{z}_3 + \bar{z}z_3)$

Hence, $AP^2 + BP^2 + CP^2 = 3|z|^2 + |z_1|^2 + |z_2|^2 + |z_3|^2 - z(\bar{z}_1 + \bar{z}_2 + \bar{z}_3) - \bar{z}(z_1 + z_2 + z_3)$

$= 1 + 3.\frac{4}{3} - z(0) - \bar{z}(0) = 5$

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13.2 n^{th} Roots Of Unity :

Let $n \in \mathbb{N}, n > 1$.

The nth roots of unity are roots of equation $z^n = 1$.

Now $1 = \cos 0 + i\sin 0 = \cos 2k\pi + i\sin 2k\pi = e^{i2k\pi}$; $k \in \mathbb{Z}$

$\therefore z^n = 1 \Rightarrow z = e^{i\frac{2k\pi}{n}}$; $k \in \mathbb{Z}$

$\Rightarrow z = 1, e^{i\frac{2\pi}{n}}, e^{i\frac{4\pi}{n}}, e^{i\frac{6\pi}{n}}, ..., e^{i\frac{2(n-1)\pi}{n}}$

NOTE :

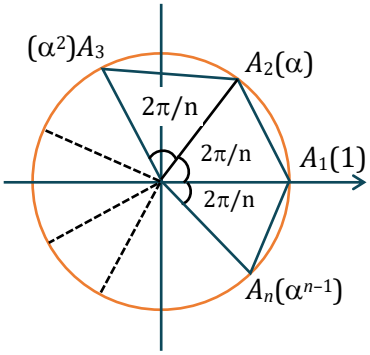
(i) Let $\alpha = e^{i\frac{2\pi}{n}}$. Then n^{th} roots of unity are $1, \alpha, \alpha^2, \alpha^3, ..., \alpha^{n-1}$

When written like this, they form a G.P. with common ratio $\alpha = e^{i\frac{2\pi}{n}}$.

Also, in this case their amplitudes are in A.P. with common-difference $\frac{2\pi}{n}$.

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(ii) The points represented by n, n^{th} roots of unity are located at the vertices of a regular polygon of n sides inscribed in a unit circle having center at origin, one vertex being on positive real axis.



(iii) If $1,\alpha_1,\alpha_2,\alpha_3,...\alpha_{n-1}$ denote the nth roots of unity, then

Sum of roots = $1 + \alpha_1 + \alpha_2 + \alpha_3 + ... + \alpha_{n-1} = 0$

Product of roots =

$$1\cdot\alpha_1\cdot\alpha_2\cdot\alpha_3\cdot....\cdot\alpha_{n-1} = (-1)^n = \begin{cases} 1, & n = odd \\ -1, & n = even \end{cases}$$

(iv) If $1, \alpha_1, \alpha_2, \alpha_3, ...\alpha_{n-1}$ denote the nth roots of unity, then

$x^n - 1 = (x - 1)(x - \alpha_1)(x - \alpha_2)....(x - \alpha_{n-1})$

Illustration 83

If $1,\alpha_1, \alpha_2, \alpha_3, ... \alpha_{n-1}$ denote the nth roots of unity, then prove that

$$(1 + \alpha_1)(1 + \alpha_2)..... (1 + \alpha_{n-1}) = \begin{cases} 0, & \text{if } n \text{ is even} \\ 1, & \text{if } n \text{ is odd.} \end{cases}$$

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Solution:

We know that $x^n - 1 = (x - 1)(x - \alpha_1)(x - \alpha_2)....(x - \alpha_{n-1})$

Putting $x = -1$ in above equation, we get

$$\Rightarrow (-1-1)(-1-\alpha_1)(-1-\alpha_2)...(-1-\alpha_{n-1}) = (-1)^n - 1$$

$$\Rightarrow (1+\alpha_1)(1+\alpha_2)...(1+\alpha_{n-1}) = \frac{(-1)^n - 1}{(-1)^n} = \begin{cases} 0, & \text{if } n \text{ is even} \\ 1, & \text{if } n \text{ is odd.} \end{cases}$$

Illustration 84

If $1,\alpha_1, \alpha_2, \alpha_3, ... \alpha_{n-1}$ denote the nth roots of unity, then prove that $(1 - \alpha_1)(1 - \alpha_2)..... (1 - \alpha_{n-1}) = n$.

Solution:

We know that $x^n - 1 = (x - 1)(x - \alpha_1)(x - \alpha_2)....(x - \alpha_{n-1})$

$$\Rightarrow (x-\alpha_1)(x-\alpha_2)...(x-\alpha_{n-1}) = \frac{x^n - 1}{x - 1}$$

$$\Rightarrow (x-\alpha_1)(x-\alpha_2)...(x-\alpha_{n-1}) = 1 + x + x^2 + ... + x^{n-1}$$

Now, putting $x = 1$ in above equation, we get

$(1 - \alpha_1)(1 - \alpha_2)..... (1 - \alpha_{n-1}) = n$.

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Illustration 85

If $1,\alpha_1,\alpha_2,\alpha_3,...\alpha_{n-1}$ denote the nth roots of unity, then prove that

$$1^p + \alpha_1^p + \alpha_2^p + ... + \alpha_{n-1}^p = \begin{cases} 0, \text{ if } p \text{ is not an integral multiple of } n \\ n, \text{ if } p \text{ is an integral multiple of } n \end{cases}$$

Solution:

Let $\alpha_k = e^{i\frac{2k\pi}{n}}$. Then let $\alpha_k = \left(e^{i\frac{2\pi}{n}}\right)^k = (\lambda)^k$, where $\lambda = e^{i\frac{2\pi}{n}}$

Then

$$(1)^p + (\alpha_1)^p + (\alpha_2)^p + ... + (\alpha_{n-1})^p = (1)^p + (\lambda)^p + (\lambda^2)^p + ... + (\lambda^{n-1})^p$$

$$= (\lambda)^0 + (\lambda)^{2p} + (\lambda)^{3p} + ... + (\lambda)^{p(n-1)} = \begin{cases} p, & \text{if } \lambda^p = 1 \\ \dfrac{(\lambda^p)^n - 1}{\lambda^p - 1}, & \text{if } \lambda^p \neq 1 \end{cases}$$

$$= \begin{cases} p, & \text{if } \lambda^p = 1 \\ 0, & \text{if } \lambda^p \neq 1 \end{cases} \quad \left[\because \quad (\lambda^p)^n = (\lambda^n)^p = \left(e^{i(2\pi)}\right)^p = 1\right]$$

$$= \begin{cases} 0, \text{ if } p \text{ is not an integral multiple of } n \\ n, \text{ if } p \text{ is an integral multiple of } n \end{cases}$$

Illustration 86

If roots of the equation $x^6 - 1 = 0$ are given by α_i where $i \in \{1, 2, 3, ..., 6\}$, then which of the following option is/are correct.

(A) $\prod_{i=1}^{6}(2-\alpha_i) = 63$ (B) $\sum_{i=1}^{6}(2-\alpha_i) = 12$ (C) $\prod_{i=1}^{6}(2-\alpha_i) = 31$ (D) $\prod_{i=1}^{6}\alpha_i = -1$

Ans. (A,B,D)

Solution:

$(x^6-1) = (x-\alpha_1)(x-\alpha_2)(x-\alpha_3)\ldots.(x-\alpha_6)$

Now put $x = 2$

$2^6-1 = (2-\alpha_1)(2-\alpha_2).....(2-\alpha_6)$

$$63 = \prod_{i=1}^{6}(2-\alpha_i)$$

$$\sum_{i=1}^{6}(2-\alpha_i) = (2-\alpha_1) + (2-\alpha_2) ..... + (2-\alpha_6) = 12 - \sum_{i=1}^{6}\alpha_i \qquad \text{[sum of roots = 0]}$$

$= 12 - 0 = 12$

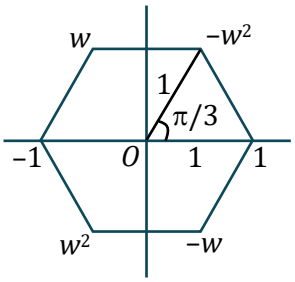
$\prod_{i=1}^{6}\alpha_i$ = product of roots = -1

Illustration 87

If z_k are the root of $z^6 = 1$, $k = 1, 2, 3,..., 6$, then find the area of the figure formed by joining the points represented by z_k.

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Solution:



$z^6 = 1$

$\Rightarrow z^3 = 1$ or $z^3 = -1$

$\Rightarrow z = 1, w, w^2$ or $-1, -w, -w^2$

Area = 6 × area(ΔOAB) = $6 \times \frac{\sqrt{3}}{4} = \frac{3\sqrt{3}}{2}$ sq. units.

Illustration 88

Find the value $\sum_{k=1}^{6}\left(\sin\frac{2\pi k}{7} - \cos\frac{2\pi k}{7}\right)$ using properties of 7th roots of unity.

Solution:

7th roots of unity are $e^{i\frac{2k\pi}{7}}; k = 0,1,2,...6$

We know that sum of 7th roots of unity = 0

$$\Rightarrow \sum_{k=0}^{6} e^{i\frac{2k\pi}{7}} = 0 \Rightarrow \sum_{k=0}^{6}\cos\frac{2k\pi}{7} + i\sin\frac{2k\pi}{7} = 0 + 0i;$$

$$\Rightarrow \sum_{k=0}^{6}\cos\frac{2k\pi}{7} = 0 \text{ and } \sum_{k=0}^{6}\sin\frac{2k\pi}{7} = 0 \Rightarrow \sum_{k=1}^{6}\cos\frac{2k\pi}{7} = -1 \text{ and } \sum_{k=1}^{6}\sin\frac{2k\pi}{7} = 0$$

Now $\sum_{k=1}^{6}\left(\sin\frac{2\pi k}{7} - \cos\frac{2\pi k}{7}\right) = \sum_{k=1}^{6}\left(\sin\frac{2\pi k}{7}\right) - \sum_{k=1}^{6}\left(\cos\frac{2\pi k}{7}\right) = 0 - (-1) = 1$

Illustration 89

Prove that $\sin\frac{\pi}{n}\sin\frac{2\pi}{n}.....\sin\frac{(n-1)\pi}{n} = \frac{n}{2^{n-1}}$; $n \in \mathbb{N}$, $n \geq 2$.

Solution:

Let n^{th} roots of unity are denoted by $\alpha_k = e^{i\frac{2k\pi}{n}}; n = 0,1,2,...n-1$

We know that $x^n - 1 = (x-1)(x-\alpha_1)(x-\alpha_2)....(x-\alpha_{n-1})$

$$\Rightarrow (x-\alpha_1)(x-\alpha_2)...(x-\alpha_{n-1}) = \frac{x^n - 1}{x-1}$$

$$\Rightarrow (x-\alpha_1)(x-\alpha_2)...(x-\alpha_{n-1}) = 1 + x + x^2 + ... + x^{n-1}$$

Now, putting $x = 1$ in above equation, we get

$(1-\alpha_1)(1-\alpha_2)..... (1-\alpha_{n-1}) = n$.

Taking mod on both sides, we get $|1-\alpha|.|1-\alpha^2|.|1-\alpha^3|....|1-\alpha^{n-1}| = n$

Using the result $\left|1-e^{i\theta}\right| = \left|2\sin\frac{\theta}{2}\right| \Rightarrow |1-\alpha_k| = \left|1-e^{i\frac{2k\pi}{n}}\right| = \left|2\sin\frac{k\pi}{n}\right|$, we get

$$\Rightarrow \left(2\sin\frac{\pi}{n}\right)\left(2\sin\frac{2\pi}{n}\right)\left(2\sin\frac{3\pi}{n}\right)...\left(2\sin\frac{(n-1)\pi}{n}\right) = n$$

$$\Rightarrow \sin\frac{\pi}{n}\sin\frac{2\pi}{n}....\sin\frac{(n-1)\pi}{n} = \frac{n}{2^{n-1}}$$

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